<u>P8100 - APPLIED REGRESSION I</u>

CONTENTS

1 INTRODUCTION

1.1 WHAT IS REGRESSION ANALYSIS?

Regression analysis refers to a collection of statistical techniques that are used to investigate the relationship between quantities in a given scientific system. For example, it can be used to answer the following questions:

- Are sons taller than their fathers?
- Does changing class size affect success of students?
- Can we predict the time of the next eruption of Mount Etna from the length of the most recent eruption?
- Do changes in diet result in changes in cholesterol level, and if so, do the results depend on other characteristics such as age, sex, and amount of exercise?
- Do countries with higher per person income have lower birth rates than countries with lower income?
- Are highway design characteristics associated with highway accident rates? Can accident rates be lowered by changing design characteristics?

- Is water usage increasing over time?

1.2 HISTORICAL OVERVIEW

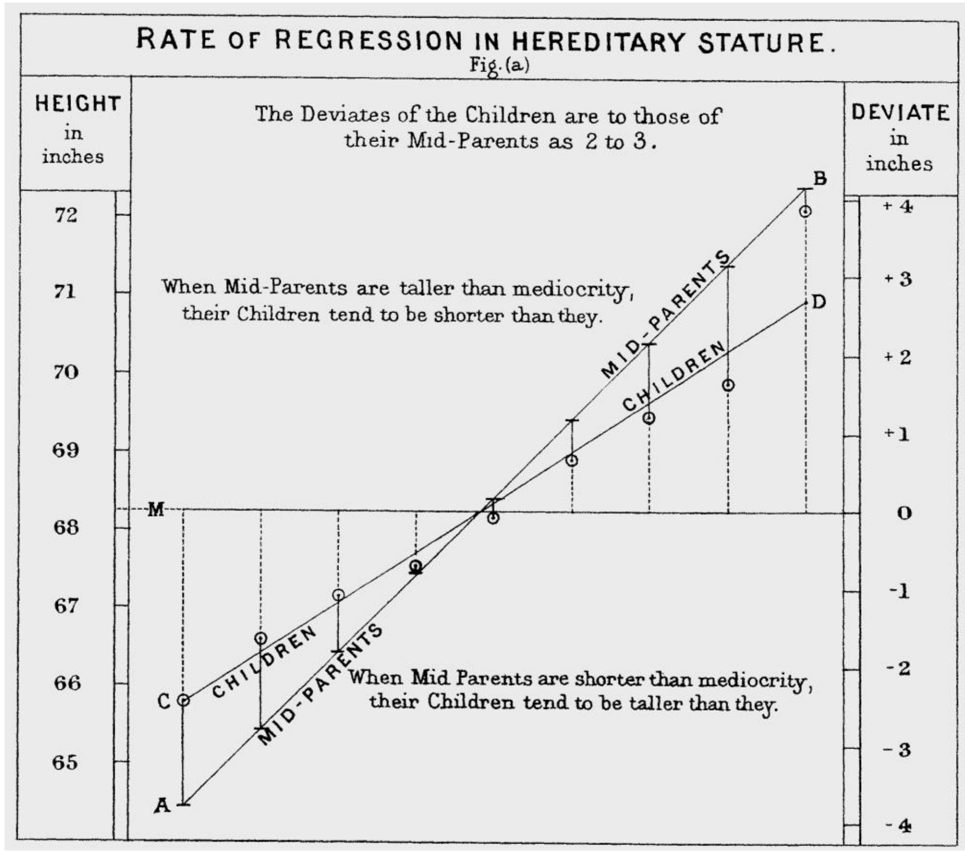
In the second half of the 19th century, Francis Galton, who was Charles Darwin's first cousin, became increasingly interested in mathematical investigations of heredity. His 1869 book, *Hereditary Genius*, was entirely devoted to the topic. Galton argued that descendants of eminent individuals were more likely to be eminent than others but to a lesser degree than their ancestors. In an 1877 paper entitled "Typical laws of heredity", Galton wished to explain why the variability of inherited characteristics (e.g. height) from an isolated human population remained constant from generation to generation. He explained the constancy through the action of two opposite forces:

- familial variability, which is variation in the characteristic arising from the different offspring of a given parental class
- reversion, which is the appearance of ancestral characteristics in an offspring

In 1895, Galton wrote his observations on sweet pea seeds as follows:

> It appeared from these experiments that the offspring did not tend to resemble their parent seeds in size, but to be always more mediocre than they were—to be smaller than the parents, if the parents were large; to be larger than the parents, if the parents were small.

Working on the heights of parents and children, Galton realized that what he had called reversion at first was a symmetric phenomenon. Not only tall parents had less tall children on average and short parents had less short children on average, but also tall children had less tall parents on average and short children had less short parents on average (see figure below)



Thereupon, Galton abandoned the term “reversion” (which was a genetic phenomenon) and adopted “regression” (which was a more a more general phenomenon). Denoting the height of mid-parents by *Y* and the height of a child by *X*, Galton wrote his law of regression as:

$$\mathrm{E}(Y \mid X) - M = \frac{2}{3}(X - M) \quad \Rightarrow \quad \mathrm{E}(Y \mid X) = \frac{2}{3}X + \frac{1}{3}M,$$

where *M* is the mean height of parents and of children. Similarly, the law of regression for children on parents is

$$\mathrm{E}(X \mid Y) - M = \frac{1}{3}(Y - M) \quad \Rightarrow \quad \mathrm{E}(X \mid Y) = \frac{1}{3}Y + \frac{2}{3}M,$$

Note that a regression slope of 2/3 for parents on children does NOT imply a regression slope of 3/2 for children on parents.

Why does regression occur? In general, suppose a first measurement X is made on a given subject, followed by a second measurement Y. Assume X is exceptionally high. As long as X and Y are imperfectly correlated, X can be thought to be made up of two components:

1. A first component that is unusually extreme and is expected to remain extreme
2. A second component that is not extreme and is expected to remain near the center of the distribution

The first measurement X is extreme because both of the components are high. However, for the second measurement Y, the first component is expected to remain high but the second component is expected to be near the center. Hence, the average value of Y will be less extreme than X and closer to the center of the distribution.

2 SIMPLE LINEAR REGRESSION

2.1 THE SIMPLE LINEAR REGRESSION MODEL

The simple linear regression model is written as

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where

$$\mathrm{E}(\varepsilon) = 0 \quad \text{so that} \quad \mathrm{E}(Y \mid X) = \beta_0 + \beta_1 X$$

and

$$\operatorname{var}(Y \mid X) = \operatorname{var}\varepsilon = \sigma^2_{Y|X} = \sigma^2$$

In the above, Y is the response (or dependent) variable, β_0 and β_1 are the intercept and slope respectively , X is the regressor (or independent) variable and ε is the model error. Since the data are collected in pairs, i.e. in the form $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$, the model above is usually written as

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, 2, \ldots, n \qquad \textbf{2-1}$$

Notes:

(a) The parameter β_0 represents the mean value of *Y* when *X* =0 and the parameter β_1 represents the change in the mean value of *Y* for each unit change in *x* (assuming *x* is continuous).
(b) Model (2-1) represents the regression of *Y* on *X*. The situation can be reversed and *X* regressed on *y*, resulting in the model:

$$X_i = \gamma_0 + \gamma_1 Y_i + \eta_i, \quad i = 1, 2, ..., n$$

(c) Model (2-1) is called a *simple* linear regression model because only one regressor is involved. It is said to be *linear* since it is linear in the β's.

2.2 THE METHOD OF LEAST-SQUARES

One method of keeping the errors as small as possible is to minimize the sum of squares of the errors and this results in ordinary least-squares (OLS) estimators of β_0 and β_1 . The sum of the squares of the errors is

$$S = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 X_i\right)^2$$

Therefore,

$$\frac{\partial S}{\partial \beta_0} = -2\sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 X_i\right)$$
$$\frac{\partial S}{\partial \beta_1} = -2\sum_{i=1}^{n} X_i \left(Y_i - \beta_0 - \beta_1 X_i\right)$$

At a minimum, both of the above partial derivatives are zero, so that

$$\sum_{i=1}^{n} Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^{n} X_i = 0 \quad \Rightarrow \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \qquad \textbf{2-2}$$

$$\sum_{i=1}^{n} X_i Y_i - \hat{\beta}_0 \sum_{i=1}^{n} X_i - \hat{\beta}_1 \sum_{i=1}^{n} X_i^2 = 0 \quad \Rightarrow \quad \sum_{i=1}^{n} X_i Y_i - n\hat{\beta}_0 \bar{X} - \hat{\beta}_1 \sum_{i=1}^{n} X_i^2 = 0 \qquad \textbf{2-3}$$

Substituting for $\hat{\beta}_0$ from Eq. (2-2) into Eq. (2-3),

$$\sum_{i=1}^{n} X_i Y_i - n\bar{X}\left(\bar{Y} - \hat{\beta}_1 \bar{X}\right) - \hat{\beta}_1 \sum_{i=1}^{n} X_i^2 = 0 \quad \Rightarrow \quad \hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} Y_i^2 - n\bar{X}^2}$$

The above can also be written as $\hat{\beta}_1 = \sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) / \sum_{i=1}^{n}(X_i - \bar{X})^2$. The least-squares estimators of β_1 and β_0 are therefore

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}, \qquad \textbf{2-4}$$
$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

The least-squares line (or <u>line of best fit</u>) is therefore

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X \qquad \textbf{2-5}$$

with $\hat{\beta}_1, \hat{\beta}_0$ given in (2-4). $\hat{\beta}_0$ is the estimated intercept and represents the estimated mean value of *Y* when *X* = 0 and $\hat{\beta}_1$ is the estimated slope and represents the estimated change in the mean value of *Y* for each unit change in *X* (assuming *X* is continuous).

Example 2.1 In 1857, the Scottish physicist James Forbes (1809-1868) investigated the relationship between atmospheric pressure (inHg) and boiling point of water (ºF) by collecting data in the Alps and in Scotland. See the table below.

observation	bp	pres
1	194.5	20.79
2	194.3	20.79
3	197.9	22.4
4	198.4	22.67
5	199.4	23.15
6	199.9	23.35
7	200.9	23.89
8	201.1	23.99

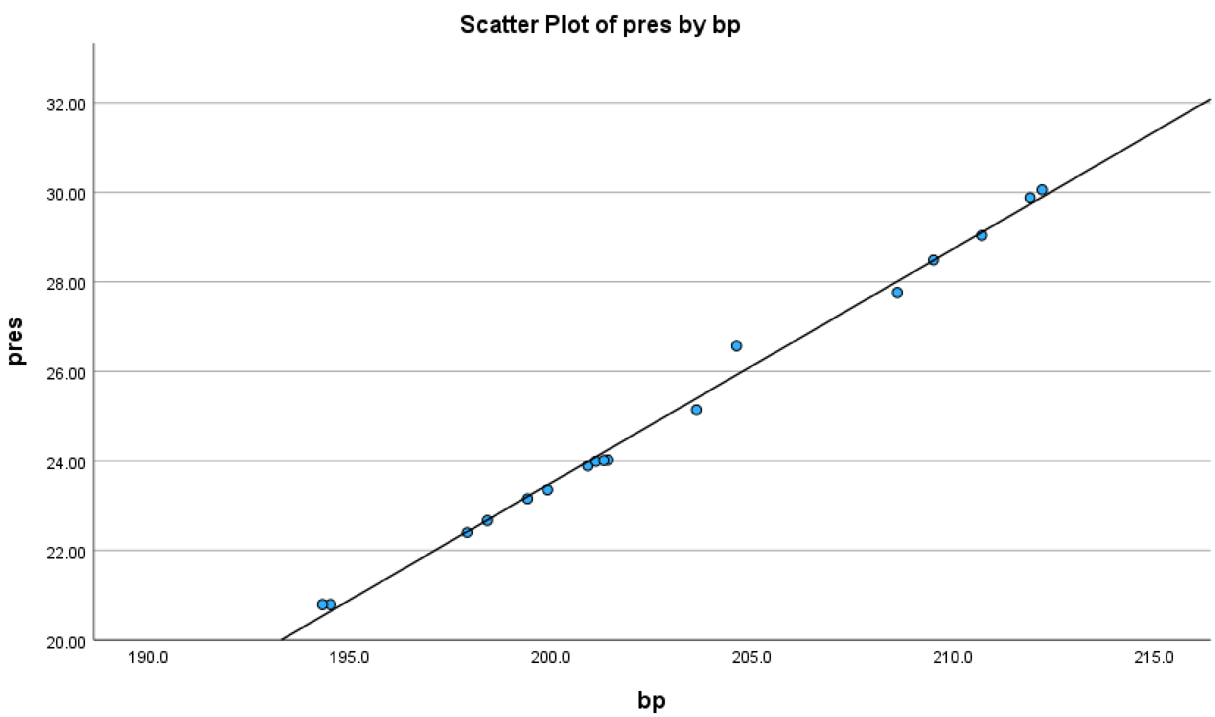
9	201.4	24.02
10	201.3	24.01
11	203.6	25.14
12	204.6	26.57
13	209.5	28.49
14	208.6	27.76
15	210.7	29.04
16	211.9	29.88
17	212.2	30.06

Since barometers at that time were fragile, regression analysis can be used to infer the value of atmospheric pressure from a simpler measurement of the boiling point of water.

(a) Obtain the line of best fit.
(b) Find the predicted value of atmospheric pressure for an altitude where the boiling point of water is 205.0 ºF.

Solution

(a) Since the objective here is to predict atmospheric pressure from boiling point of water, the first variable is the response (*Y*) and the second variable is the regressor (*X*). We start with a scatter plot of *Y vs. X*.



We have

$$\sum x_i y_i = 86{,}774.77, \quad \bar{x} = 202.95, \quad \bar{y} = 25.06$$
$$\sum x_i^2 = 700{,}553.91$$

The least-squares estimates are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2} = .52$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = -81.06$$

The line of best fit therefore is

$$\hat{y} = -81.06 + .52x$$

(b) For x= 205.0, the predicted value of the atmospheric pressure is
$$\hat{y} = -81.06 + .52(205.0) = 25.54 \text{ inHg}$$

Notes

(i) Rounding off errors imply that the estimates above will be slightly different from those obtained from a software.
(ii) For reasons to be discussed later, Forbes actually used the log to base 10 of the atmospheric pressures. His line of best fit was therefore different.

Using $\hat{\beta}_0$ in (2-4), the line of best fit in (2-5) can also be written as

$$\hat{Y} = \bar{Y} - \hat{\beta}_1 \bar{X} + \hat{\beta}_1 X = \bar{Y} + \hat{\beta}_1 \left(X - \bar{X} \right) \qquad \textbf{2-6}$$

This implies that if the centered values of *x* were used, i.e. $X = X - \bar{X}$, then

$$\hat{Y} = \bar{Y} + \hat{\beta}_1 X$$

i.e. a regression of Y on X has intercept $\overline{Y}$ and slope $\hat{\beta}_1$. The above is known as the centered model.

Example 2.2

Obtain the centered model for the Forbes data in Example 2.1

Solution

For the Forbes data, we have

$$\overline{y} = 25.06, \hat{\beta}_1 = .52 \text{ and } \overline{x} = 202.95,$$

The centered model is

$$\hat{Y} = \overline{Y} + \hat{\beta}_1 (x - \overline{x}) = 25.06 + .52(x - 202.95) = 25.06 + .52\tilde{x},$$

where $\tilde{x} = x - 202.95$ is the centered value of *x*.

It is also possible to calculate the residual $\hat{\varepsilon}_i$ associated with each observation, where

$$\hat{\varepsilon}_i = Y_i - \hat{Y}_i.$$

The residuals are an important component of any regression analysis. Note that:

$$\begin{aligned}
\sum_{i=1}^{n} \hat{\varepsilon}_i &= \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right) \\
&= \sum_{i=1}^{n} \left[Y_i - \left(\overline{Y} + \hat{\beta}_1 \{x_i - \overline{x}\} \right) \right] \\
&= \sum_{i=1}^{n} \left(Y_i - \overline{Y}\right) - \hat{\beta}_1 \sum_{i=1}^{n} \left(x_i - \overline{x}\right) \\
&= 0 - \hat{\beta}_1 (0) \\
&= 0
\end{aligned}$$

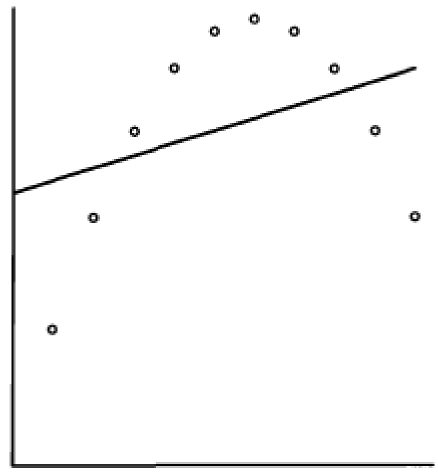
2.3 THE GAUSS MARKOV CONDITIONS

The least-squares method gives good predictions when the following conditions are met:

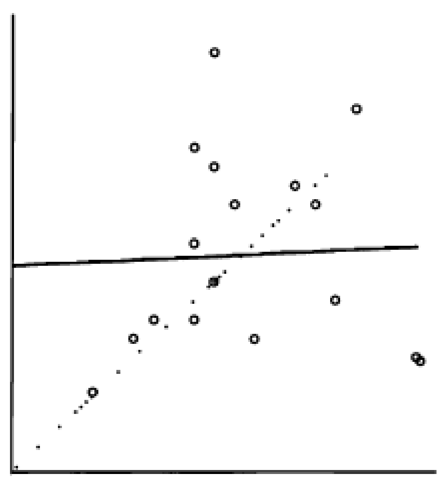
(a) The figure below shows that the least-squares method is not appropriate. It suggests the following condition should be met for good prediction:

$$C_1: \qquad \mathrm{E}(\varepsilon_i) = 0$$

This implies that, for the linear regression model in (2-2), $\mathrm{E}(Y_i) = \beta_0 + \beta_1 x_i$.



(b) Suppose the true regression line is shown by the dotted line below:



Because there is an increase in $\text{var}(\varepsilon_i)$ with x_i (i.e. there is heteroscedasticity), the line of best fit shown by the continuous line represents a bad fit. It therefore seems reasonable to impose the condition:

$$C_2: \quad \text{var}(\varepsilon_i) = \sigma^2, \ \text{ a constant}$$

This implies that, for the linear regression model in (2-2), $\text{var}(Y_i) = \sigma^2$.

(c) Suppose we have the two observations (x_1, y_1) and (x_2, y_2), and we make 10 copies of each. Are we much better off with the 20 'observations' than the original 2. The answer is no and suggests a third reasonable:

$$C_3: \quad \text{corr}(\varepsilon_i, \varepsilon_i) = 0$$

The conditions C1-C3 are known as the Gauss-Markov conditions.

2.4 REGRESSION AND CORRELATION

In simple linear regression, the sample correlation coefficient between *X* and *Y* is defined by

$$r_{XY} = \frac{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)\left(Y_i - \bar{Y}\right)}{\sqrt{\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2}} = \frac{S_{XY}}{\sqrt{S_{XX} S_{YY}}}$$

Since $\hat{\beta}_1 = S_{XY} / S_{XX}$, we also have

$$r_{XY} = \hat{\beta}_1 \sqrt{\frac{S_{XX}}{S_{YY}}} = \hat{\beta}_1 \sqrt{\frac{\text{var}X}{\text{var}Y}}$$

If we note the population correlation coefficient by ρ and the population regression slope by β_1, a similar relationship holds:

$$\rho = \beta_1 \frac{\sigma_X}{\sigma_Y}.$$

It is seen that $\rho = 0$ iff $\beta_1 = 0$. Therefore, a test of $H_0 : \beta_1 = 0$ (no regression) is equivalent to a test of $H_0 : \rho = 0$ (no correlation), i.e. both tests are testing for the same thing and yield identical p-values. For the $H_0 : \beta_1 = 0$, we use

$$t_{test} = \frac{\hat{\beta}_1}{\mathrm{SE}\hat{\beta}_1} \sim t_{n-2}$$

For $H_0 : \rho = 0$, we use

$$t_{test} = \frac{r}{\sqrt{1-r^2}} \sqrt{n-2} \sim t_{n-2}$$

For $H_0 : \rho = \rho_0 \;\; (\rho_0 \neq 0)$, we use

$$z_{test} = \frac{\frac{1}{2}\left\{ \ln\left(\frac{1+r}{1-r}\right) - \ln\left(\frac{1+\rho_0}{1-\rho_0}\right) \right\}}{1/\sqrt{n-3}} \sim N(0,\, 1)$$

2.5 PROPERTIES OF ESTIMATORS

From Eq. (2-4),

$$
\begin{aligned}
\mathrm{E}\hat{\beta}_1 &= \mathrm{E}\,\frac{\sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y}}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2} \\
&= \frac{1}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}\,\mathrm{E}\left(\sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y}\right) \\
&= \frac{1}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}\left(\sum_{i=1}^{n} x_i \mathrm{E}Y_i - n\bar{x}\mathrm{E}\bar{Y}\right) \\
&= \frac{1}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}\left\{\sum_{i=1}^{n} x_i \mathrm{E}\left(\beta_0 + \beta_1 x_i + \varepsilon_i\right) - n\bar{x}\mathrm{E}\left(\beta_0 + \beta_1\bar{x} + \bar{\varepsilon}\right)\right\} \\
&= \frac{1}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}\left(\beta_0\sum_{i=1}^{n} x_i + \beta_1\sum_{i=1}^{n} x_i^2 - \beta_0\sum_{i=1}^{n} x_i - n\beta_1\bar{x}^2\right) \\
&= \frac{1}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}\left\{\beta_1\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)\right\} \\
&= \beta_1
\end{aligned}
\qquad \textbf{2-7}
$$

and

$$
\begin{aligned}
\mathrm{E}\hat{\beta}_0 &= \mathrm{E}\left(\bar{Y} - \hat{\beta}_1\bar{x}\right) \\
&= \mathrm{E}\left(\beta_0 + \beta_1\bar{x} + \bar{\varepsilon} - \hat{\beta}_1\bar{x}\right) \\
&= \beta_0 + \beta_1\bar{x} - \bar{x}\mathrm{E}\hat{\beta}_1 \\
&= \beta_0 + \beta_1\bar{x} - \beta_1\bar{x} \\
&= \beta_0
\end{aligned}
\qquad \textbf{2-8}
$$

Both least-squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are therefore said to be *unbiased.*

Similarly, it can be shown that

$$\operatorname{var}\hat{\beta}_1 = \frac{\sigma^2}{S_{xx}},$$
$$\operatorname{var}\hat{\beta}_0 = \sigma^2\left(\frac{1}{n} + \frac{\overline{x}^2}{S_{xx}}\right), \qquad \textbf{2-9}$$

where $S_{xx} = \sum_i (x_i - \overline{x})^2$.

One difficulty with both of the equations above is that the variance σ^2 $(= \operatorname{var}\varepsilon_i)$ is an unknown parameter. Using the notation $S_{yy} = \sum_i (y_i - \overline{y})^2$ and $S_{xy} = \sum_i (x_i - \overline{x})(y_i - \overline{y})$, an unbiased estimator of σ^2 is

$$\begin{aligned}
s^2 &= \frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^2}{n-2} \\
&= \frac{\sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2}{2} \\
&= \frac{\sum_{i=1}^{n} \left[Y_i - \left\{\overline{Y} + \hat{\beta}_1\left(x_i - \overline{x}\right)\right\}\right]^2}{n-2} \\
&= \frac{\sum_{i=1}^{n} \left\{\left(Y_i - \overline{Y}\right) - \hat{\beta}_1\left(x_i - \overline{x}\right)\right\}^2}{n-2} \\
&= \frac{\sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)^2 + \hat{\beta}_1^2 \sum_{i=1}^{n} \left(x_i - \overline{x}\right)^2 - 2\hat{\beta}_1 \sum_{i=1}^{n} \left(Y_i - \overline{Y}\right)\left(x_i - \overline{x}\right)}{n-2} \\
&= \frac{S_{yy} + \frac{S_{xy}^2}{S_{xx}^2} S_{xx} - 2\frac{S_{xy}}{S_{xx}} S_{xy}}{n-2}
\end{aligned}$$

$$\therefore \quad s^2 = \hat{\sigma}^2 = \frac{S_{yy} - S_{xy}^2 / S_{xx}}{n-2} \qquad \textbf{2-10}$$

(The 2 in the denominator is because 2 degrees of freedom are lost when $\hat{\beta}_0$ and $\hat{\beta}_1$, therefore s^2 is associated with *n* -2 d.f.) The variances in (2-9) are then estimated by the squares of the following standard errors:

$$\mathrm{SE}\hat{\beta}_1 = s\sqrt{\frac{1}{S_{xx}}},$$
$$\mathrm{SE}\hat{\beta}_0 = s\sqrt{\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}}. \qquad \textbf{2-11}$$

If the additional assumption on the errors ε_i is made that they are each normally distributed, i.e. $\varepsilon_i \sim \mathrm{N}(0,\ \sigma^2)$, then $Y_i \sim \mathrm{N}(\beta_0 + \beta_1 x_i,\ \sigma^2)$. But Eq. (2-4) shows that each $\hat{\beta}_i$ is a linear combination of the Y_i's, so that the $\hat{\beta}_i$'s are also normally distributed. With the expectations in (2-7) and (2-8) and the standard errors in (2-11), this implies that

$$\frac{\hat{\beta}_1 - \beta_1}{\mathrm{SE}\hat{\beta}_1} \sim t_{n-2}, \qquad \frac{\hat{\beta}_0 - \beta_0}{\mathrm{SE}\hat{\beta}_0} \sim t_{n-2}, \qquad \textbf{2-12}$$

where t_{n-2} is the student's *t*-distribution with (*n* - 2) degrees of freedom. The above can be used to test the hypothesis $H_0 : \beta_i = 0$ vs. $H_0 : \beta_i \neq 0$.

Example 2.3

For the Forbes data in Example 2.1, the following are obtained from a standard software.

Coefficients

Model		B	Std. Error
1	(Constant)	-81.064	2.052
	bp	.523	.010

Calculate the p-values for the hypotheses that: (i) $H_0 : \beta_1 = 0$ vs. $H_0 : \beta_1 \neq 0$, (ii) $H_0 : \beta_0 = 0$ vs. $H_0 : \beta_0 \neq 0$

Solution

(i) For $H_0 : \beta_1 = 0$ vs. $H_0 : \beta_1 \neq 0$, we have, with $n = 17$,

$$t_{test} = \frac{.523 - 0}{.010} = 52.30 \sim t_{15}$$

so that, from t-tables (see below),

$$p\text{val=2Pr}\left\{t_{15} > 52.3\right\} < .001$$

Hence, the null hypothesis is rejected at the 5% level.

(ii) Similarly, for $H_0 : \beta_0 = 0$ vs. $H_0 : \beta_0 \neq 0$, we have,

$$t_{test} = \frac{-81.064 - 0}{2.052} = -39.50 \sim t_{15}$$

so that, from t-tables,

$$p_{\text{val}}\text{=2Pr}\left\{t_{15} < -39.50\right\} < .001$$

Hence the null hypothesis is rejected at the 5% level.

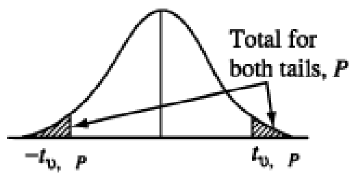


Table A3 **Percentage points of the t distribution**

The function tabulated is $t_{\nu,P}$, the value exceeded in both directions with probability P in a t distribution with ν degrees of freedom (the $100P$ percentage point).

Degrees of freedom, ν	Probability of greater value, P												
	0·9	0·8	0·7	0·6	0·5	0·4	0·3	0·2	0·1	0·05	0·02	0·01	0·001
1	0·158	0·325	0·510	0·727	1·000	1·376	1·963	3·078	6·314	12·706	31·821	63·657	636·619
2	0·142	0·289	0·445	0·617	0·816	1·061	1·386	1·886	2·920	4·303	6·965	9·925	31·598
3	0·137	0·277	0·424	0·584	0·765	0·978	1·250	1·638	2·353	3·182	4·541	5·841	12·924
4	0·134	0·271	0·414	0·569	0·741	0·941	1·190	1·533	2·132	2·776	3·747	4·604	8·610
5	0·132	0·267	0·408	0·559	0·727	0·920	1·156	1·476	2·015	2·571	3·365	4·032	6·869
6	0·131	0·265	0·404	0·553	0·718	0·906	1·134	1·440	1·943	2·447	3·143	3·707	5·959
7	0·130	0·263	0·402	0·549	0·711	0·896	1·119	1·415	1·895	2·365	2·998	3·499	5·408
8	0·130	0·262	0·399	0·546	0·706	0·889	1·108	1·397	1·860	2·306	2·896	3·355	5·041
9	0·129	0·261	0·398	0·543	0·703	0·883	1·100	1·383	1·833	2·262	2·821	3·250	4·781
10	0·129	0·260	0·397	0·542	0·700	0·879	1·093	1·372	1·812	2·228	2·764	3·169	4·587
11	0·129	0·260	0·396	0·540	0·697	0·876	1·088	1·363	1·796	2·201	2·718	3·106	4·437
12	0·128	0·259	0·395	0·539	0·695	0·873	1·083	1·356	1·782	2·179	2·681	3·055	4·318
13	0·128	0·259	0·394	0·538	0·694	0·870	1·079	1·350	1·771	2·160	2·650	3·012	4·221
14	0·128	0·258	0·393	0·537	0·692	0·868	1·076	1·345	1·761	2·145	2·624	2·977	4·140
15	0·128	0·258	0·393	0·536	0·691	0·866	1·074	1·341	1·753	2·131	2·602	2·947	4·073

The relationships in (2-12) can also be used to build $100(1-\alpha)\%$ confidence intervals (C.I.'s) for β_i:

$$\hat{\beta}_j - t_{n-2;\alpha/2}\mathrm{SE}\hat{\beta}_j < \beta_j < \hat{\beta}_j + t_{n-2;\alpha/2}\mathrm{SE}\hat{\beta}_j,$$

where $t_{n-2;\alpha/2}$ is the upper $\alpha/2$ point of the t_{n-2} distribution.

Example 2.4

For the Forbes data in Example 2.1, construct 95% C.I. for β_0 and β_1.

Solution

From t-tables, $t_{15;.025} = 2.131$ so that a 95% C.I. for β_1 is

$$\begin{aligned} &.523-(2.131)(.010) < \beta_1 < .523+(2.131)(.010) \\ &\text{i.e.,} \qquad .502 < \beta_1 < .544 \end{aligned}$$

and a 95% C.I. for β_0 is

$$\begin{aligned} &-81.064-(2.131)(2.052) < \beta_0 < -81.064-(2.131)(2.052) \\ &\text{i.e.,} \qquad -85.437 < \beta_0 < -76.691 \end{aligned}$$

2.6 USING REGRESSION TO MAKE PREDICTIONS

Suppose we wish to predict the value of *y* for a new value of *x*, namely x_0. Using (2.5), the predicted value of *y* is

$$\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

This means that

$$\begin{aligned} \mathrm{E}\hat{Y}_0 &= \mathrm{E}\hat{\beta}_0 + \mathrm{E}\hat{\beta}_1 x_0 \\ &= \beta_0 + x_0 \mathrm{E}\hat{\beta}_1 \\ &= \beta_0 + \beta_1 x_0 \end{aligned}$$

and

$$\begin{aligned}\operatorname{var}\hat{Y}_0 &= \operatorname{var}\left\{\bar{Y} + \hat{\beta}_1\left(x_0 - \bar{x}\right)\right\} \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2}{S_{xx}}\left(x_0 - \bar{x}\right)^2 \\ &= \sigma^2\left\{\frac{1}{n} + \frac{\left(x_0 - \bar{x}\right)^2}{S_{xx}}\right\}\end{aligned} \qquad \textbf{2-13}$$

Now, let Y_0 be the observation, corresponding to x_0, that would have been made if we were able to. The error in using the predicted value $\hat{Y}_0$ is

$$Y_0 - \hat{Y}_0$$

This implies that

$$\begin{aligned}\mathrm{E}\left(Y_0 - \hat{Y}_0\right) &= \mathrm{E}Y_0 - \mathrm{E}\hat{Y}_0 \\ &= \left(\beta_0 + \beta_1 x_0\right) - \left(\beta_0 + \beta_1 x_0\right) \\ &= 0\end{aligned}$$

and

$$\begin{aligned}\operatorname{var}\left(Y_0 - \hat{Y}_0\right) &= \operatorname{var} Y_0 + \operatorname{var}\hat{Y}_0 \\ &= \sigma^2 + \sigma^2\left\{\frac{1}{n} + \frac{\left(x_0 - \bar{x}\right)^2}{S_{xx}}\right\} \\ &= \sigma^2\left\{1 + \frac{1}{n} + \frac{\left(x_0 - \bar{x}\right)^2}{S_{xx}}\right\}\end{aligned} \qquad \textbf{2-14}$$

Therefore, a $100(1-\alpha)\%$ prediction interval *on a future observation at* $x = x_0$ is

$$\left[\hat{Y}_0 - t_{n-2;\alpha/2}\, s\sqrt{1 + \frac{1}{n} + \frac{\left(x_0 - \bar{x}\right)^2}{S_{xx}}},\ \hat{Y}_0 + t_{n-2;\alpha/2}\, s\sqrt{1 + \frac{1}{n} + \frac{\left(x_0 - \bar{x}\right)^2}{S_{xx}}}\right] \qquad \textbf{2-15}$$

On the other hand, a $100(1-\alpha)\%$ C.I. *on the mean value of Y at* $x = x_0$ is

$$\left[\hat{Y}_0 - t_{n-2;\alpha/2} s\sqrt{\frac{1}{n}+\frac{\left(x_0-\bar{x}\right)^2}{S_{xx}}}, \hat{Y}_0 + t_{n-2;\alpha/2} s\sqrt{\frac{1}{n}+\frac{\left(x_0-\bar{x}\right)^2}{S_{xx}}}\right] \qquad \textbf{2-16}$$

Example 2.5

In Example 2.1, find a 95% prediction interval on a future observation made at $x_0 = 205.0\ {}^0\mathrm{F}$

Solution

The predicted value at $x_0 = 205.0$ is $\hat{Y}_0 = 25.54$. Moreover,

$$\begin{aligned}
S_{xy} &= \sum_i \left(x_i - \bar{x}\right)\left(Y_i - \bar{Y}\right) \\
&= \sum_i x_i Y_i - n\bar{x}\bar{Y} \\
&= 277.54, \\
S_{xx} &= \sum_i \left(x_i - \bar{x}\right)^2 \\
&= \sum_i x_i^2 - n\bar{x}^2 \\
&= 530.78, \\
S_{yy} &= \sum_i \left(y_i - \bar{y}\right)^2 \\
&= \sum_i y_i^2 - n\bar{y}^2 \\
&= 145.94
\end{aligned}$$

and

$$s = \sqrt{\frac{S_{yy} - S_{xy}^2 / S_{xx}}{n-2}} = .23$$

Therefore, a 95% prediction interval on a future interval at $x = x_0$ is

$$\left[\hat{Y}_0 - t_{n-2;\alpha/2}s\sqrt{1+\frac{1}{n}+\frac{\left(x_0-\bar{x}\right)^2}{S_{xx}}}, \hat{Y}_0 + t_{n-2;\alpha/2}s\sqrt{1+\frac{1}{n}+\frac{\left(x_0-\bar{x}\right)^2}{S_{xx}}}\right]$$

$$=\left[25.54-(2.131)(.23)\sqrt{1+\frac{1}{17}+\frac{(205.0-202.95)^2}{530.78}},\right.$$

$$\left.25.54+(2.131)(.23)\sqrt{1+\frac{1}{17}+\frac{(205.0-202.95)^2}{530.78}}\right]$$

$$=(25.03, 26.05)$$

2.7 PARTITIONING OF TOTAL VARIABILITY: ANALYSIS OF VARIANCE

The regression of *Y* on *x* can be viewed as *x* trying to explain as much variation in *Y* as possible, with the unexplained variation representing the residual error. Writing

$$\left(Y_i-\bar{Y}\right)\equiv\left(\hat{Y}_i-\bar{Y}\right)+\left(Y_i-\hat{Y}_i\right)$$

we have

$$\sum_{i=1}^{n}\left(Y_i-\bar{Y}\right)^2=\sum_{i=1}^{n}\left(\hat{Y}_i-\bar{Y}\right)^2+\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2+2\sum_{i=1}^{n}\left(\hat{Y}_i-\bar{Y}\right)\left(Y_i-\hat{Y}_i\right)$$

The cross-product term can be shown to be zero, so that

$$\underbrace{\sum_{i=1}^{n}\left(Y_i-\bar{Y}\right)^2}_{\text{SS}_{\text{total}}}=\underbrace{\sum_{i=1}^{n}\left(\hat{Y}_i-\bar{Y}\right)^2}_{\text{SS}_{\text{reg}}}+\underbrace{\sum_{i=1}^{n}\left(Y_i-\hat{Y}_i\right)^2}_{\text{SS}_{\text{res}}} \qquad \textbf{2-17}$$

Furthermore, under normal theory assumptions,

$$\frac{\text{SS}_{\text{res}}}{\sigma^2}\sim\chi^2_{n-2}$$

Under $H_0:\beta_1=0$,

$$\frac{\text{SS}_{\text{reg}}}{\sigma^2}\sim\chi^2_1$$

Therefore, under $H_0:\beta_1=0$,

$$F_{test} = \frac{\mathrm{SS}_{\mathrm{reg}} / 1}{\mathrm{SS}_{\mathrm{res}} / (n-2)} = \frac{\mathrm{MS}_{\mathrm{reg}}}{s^2} \sim F_{1,n-2}, \tag{2-18}$$

where $F_{1,n-2}$ is the *F*-distribution with 1 numerator d.f. and $n-2$ denominator d.f. The p-value for testing $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$ is $p_{\mathrm{val}} = \Pr\{F_{1,n-2} > F_{test}\}$.

In the equation above, $\mathrm{SS}_{\mathrm{res}}$ can be calculated from Eq. (2-10):

$$\mathrm{SS}_{\mathrm{res}} = S_{yy} - S_{xy}^2 / S_{xx}.$$

To calculate $\mathrm{SS}_{\mathrm{reg}}$, we note from Eq. (2-17) that

$$S_{yy} = \mathrm{SS}_{\mathrm{reg}} + \left(S_{yy} - \frac{S_{xy}^2}{S_{xx}} \right) \quad \Rightarrow \quad \boxed{\mathrm{SS}_{\mathrm{reg}} = \frac{S_{xy}^2}{S_{xx}}}$$

The analysis of variance (ANOVA) table looks as follows:

source	Sum of Squares (SS)	df	Mean Squares (MS)	F
regression	SS_{reg}	1	$SS_{\mathrm{reg}} / 1$	
residual	SS_{res}	*n - 2*	$s^2 = SS_{\mathrm{res}} / (n-2)$	$F = \frac{SS_{\mathrm{reg}}}{s^2}$
total	SS_{total}	*n - 1*		

Example 2.6

Use the sums of squares in Example 2.4 to construct an ANOVA table for the Forbes data and calculate the *p*-value for the hypothesis $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$.

Solution

From Example 2.4, we have $S_{xx} = 530.78$, $S_{yy} = 145.94$ and $S_{xy} = 277.54$. Then,

$$s^2 = \frac{S_{yy} - S_{xy}^2 / S_{xx}}{n-2} = .054 .$$

Moreover,

$$\text{SS}_{\text{reg}} = \frac{S_{xy}^2}{S_{xx}} = 145.12$$

Therefore,

$$F_{test} = \frac{\text{SS}_{\text{reg}} / 1}{s^2} = 2677.11 \sim F_{1,15}$$

From the F-tables, $p_{\text{val}} = \Pr\{F_{1,15} > 2677.11\} < .01$ so that we reject $H_0 : \beta_1 = 0$ at the 5% level.

Percentage points of the F-distribution with degrees of freedom ν_1 and ν_2
Upper 5% points

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0	249.1	250.1	251.1	252.2	253.3	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
∞	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

Percentage points of the F-distribution with degrees of freedom ν_1 and ν_2
Upper 1% points

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	∞
1	4052	4999.5	5403	5625	5764	5859	5928	5981	6022	6056	6106	6157	6209	6235	6261	6287	6313	6339	6366
2	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50
3	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13
4	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46
5	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02
6	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88
7	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65
8	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86
9	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31
10	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00	3.91
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69	3.60
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45	3.36
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25	3.17
14	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09	3.00
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96	2.87
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84	2.75
17	8.40	6.11	5.18	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75	2.65
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66	2.57
19	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58	2.49
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52	2.42
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46	2.36
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40	2.31
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35	2.26
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31	2.21
25	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.54	2.45	2.36	2.27	2.17
26	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29	3.18	3.09	2.96	2.81	2.66	2.58	2.50	2.42	2.33	2.23	2.13
27	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26	3.15	3.06	2.93	2.78	2.63	2.55	2.47	2.38	2.29	2.20	2.10
28	7.64	5.45	4.57	4.07	3.75	3.53	3.36	3.23	3.12	3.03	2.90	2.75	2.60	2.52	2.44	2.35	2.26	2.17	2.06
29	7.60	5.42	4.54	4.04	3.73	3.50	3.33	3.20	3.09	3.00	2.87	2.73	2.57	2.49	2.41	2.33	2.23	2.14	2.03
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11	2.01
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	1.92	1.80
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73	1.60
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53	1.38
∞	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32	1.00

2.8 MEASURING GOODNESS-OF-FIT IN SIMPLE REGRESSION

A good fit is obtained when the residuals are small. Thus, the goodness-of-fit could be measured by the sum of the residuals, i.e. $\sum_{i=1}^{n} \hat{\varepsilon}_i^2$. Since this quantity is dependent on the units in which the y_i's are measured, a better measure of goodness-of-fit is

$$R^2 = 1 - \frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^2}{\sum_{i=1}^{n} \left(y_i - \overline{y}\right)^2} \qquad \textbf{2-19}$$

The value of R^2 lies in [0, 1], with larger values indicating better fit.

In light of our ANOVA interpretation of simple linear regression (see last section), we are now able to give a new interpretation to the quantity R^2 in Eq. (2-19). Recall that

$$\begin{aligned} R^2 &= 1 - \frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^2}{\sum_{i=1}^{n} \left(y_i - \overline{y}\right)^2} \\ &= 1 - \frac{SS_{\text{res}}}{SS_{\text{total}}} \\ &= \frac{SS_{\text{total}} - SS_{\text{res}}}{SS_{\text{total}}} \\ &= \frac{SS_{\text{reg}}}{SS_{\text{total}}} \end{aligned} \qquad \textbf{2-20}$$

R^2 is thus the proportion of the total variation in *Y* that is explained by its regression on *x.* It is also called the coefficient of determination. It can be calculated as

$$R^2 = \frac{S_{xy}^2 / S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} S_{yy}}$$

This is the square of the correlation coefficient between *Y* and *X*.

3 MULTIPLE LINEAR REGRESSION

3.1 THE MULTIPLE LINEAR REGRESSION MODEL AND THE GENERAL LINEAR MODEL

Very often, more than one regressor needs to be considered when performing a regression analysis and the simple linear regression model of Chapter 2 must be extended to a *multiple linear regression model*. For simple linear regression, with response *Y* and predictor *X* the data were of the form

$$\begin{array}{cc} Y_1 & x_1 \\ Y_2 & x_2 \\ \vdots & \vdots \\ Y_n & x_n \end{array}$$

For the *i*th observation, the simple linear regression model is written as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (i = 1, 2, ..., n)$$

In a multiple linear regression situation, with response *Y* and *k* predictors $x_1, x_2, ..., x_k$ the data are in the form

$$\begin{array}{ccccc} Y_1 & x_{11} & x_{21} & \cdots & x_{k1} \\ Y_2 & x_{12} & x_{22} & \cdots & x_{k2} \\ \vdots & \vdots & \vdots & & \vdots \\ Y_n & x_{1n} & x_{2n} & \cdots & x_{kn} \end{array}$$

For the *i*th observation, the multiple linear regression model is

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + ... + \beta_k x_{ki} + \varepsilon_i \quad (i = 1, 2, ..., n) \qquad \textbf{3-1}$$

The parameter β_0 represents the mean value of *Y* when all $x_j^{'}s$ (*j* = 1, 2, …, *k*) are set to zero. The parameter β_j is the *j*th partial regression coefficient and represents the change in the mean value of *Y* for each unit change in x_j (assuming it is continuous) while assuming that all other regressors are kept constant.

To represent all *n* observations, we need a matrix-vector representation:

$$\mathbf{Y}=\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \qquad \mathbf{X}=\begin{bmatrix} 1 & x_{11} & x_{21} & \cdots & x_{k1} \\ 1 & x_{12} & x_{22} & \cdots & x_{k2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{1n} & x_{2n} & \cdots & x_{kn} \end{bmatrix}, \qquad \boldsymbol{\beta}=\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \qquad \boldsymbol{\varepsilon}=\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

The equations in Eq. (3-1) can then be written as

$$\mathbf{Y}=\mathbf{X}\boldsymbol{\beta}+\boldsymbol{\varepsilon} \tag{3-2}$$

The model above is called the general linear model. From the Gauss-Markov condition C_1 in Sec. 2.3, it follows that $\mathrm{E}\boldsymbol{\varepsilon}=\mathbf{0}$ so that the linear model can also be written as

$$\mathrm{E}\mathbf{Y}=\mathbf{X}\boldsymbol{\beta} \tag{3-3}$$

For the Forbes data in Example 2.1, the general linear model is written as

$$\begin{bmatrix} 20.79 \\ 20.79 \\ 22.40 \\ 22.67 \\ 23.15 \\ 23.35 \\ 23.89 \\ 23.99 \\ 24.02 \\ 24.01 \\ 25.14 \\ 26.57 \\ 28.49 \\ 27.76 \\ 29.04 \\ 29.88 \\ 30.06 \end{bmatrix}=\begin{bmatrix} 1 & 194.5 \\ 1 & 194.3 \\ 1 & 197.9 \\ 1 & 198.4 \\ 1 & 199.4 \\ 1 & 199.9 \\ 1 & 200.9 \\ 1 & 201.1 \\ 1 & 201.4 \\ 1 & 201.3 \\ 1 & 203.6 \\ 1 & 204.6 \\ 1 & 209.5 \\ 1 & 208.6 \\ 1 & 210.7 \\ 1 & 211.9 \\ 1 & 212.2 \end{bmatrix}\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}+\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \\ \varepsilon_8 \\ \varepsilon_9 \\ \varepsilon_{10} \\ \varepsilon_{11} \\ \varepsilon_{12} \\ \varepsilon_{13} \\ \varepsilon_{14} \\ \varepsilon_{15} \\ \varepsilon_{16} \\ \varepsilon_{17} \end{bmatrix}$$

As another example, consider the following data which show that the yield of a chemical reaction (*Y*) as a function of temperature (x_1) and reaction time (x_2) .

Y (yield, %)	x_1 (temperature, ^{0}F)	x_2 (time, hr)
77	160	1
79	160	2
82	165	1
83	165	2
85	170	1
88	170	2
90	175	1
93	175	2

The general linear model for the data above is written as

$$\begin{bmatrix} 77 \\ 79 \\ 82 \\ 83 \\ 85 \\ 88 \\ 90 \\ 93 \end{bmatrix} = \begin{bmatrix} 1 & 160 & 1 \\ 1 & 160 & 2 \\ 1 & 165 & 1 \\ 1 & 165 & 2 \\ 1 & 170 & 1 \\ 1 & 170 & 2 \\ 1 & 175 & 1 \\ 1 & 175 & 2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \\ \varepsilon_8 \end{bmatrix} \tag{3-4}$$

3.2 LEAST SQUARES METHOD FOR MULTIPLE LINEAR REGRESSION

From the model in (3-2), the sum of squares of errors is

$$S = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^{/}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

$$\therefore \quad \frac{\partial S}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^{/}\mathbf{Y} + 2(\mathbf{X}^{/}\mathbf{X})\boldsymbol{\beta},$$

where the symbol $^{/}$ indicates the transpose of a matrix. The sum S achieves a minimum when the partial derivative is set to zero, so the least squares estimators $\hat{\boldsymbol{\beta}}$ satisfy

$$(\mathbf{X}^{/}\mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}^{/}\mathbf{Y}$$

These are known as the <u>normal equations</u>, and they have solution

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = (\mathbf{X}^{/}\mathbf{X})^{-1}\mathbf{X}^{/}\mathbf{Y} \qquad \textbf{3-5}$$

as long as the inverse of the matrix $\mathbf{X}^{/}\mathbf{X}$ exists.

The inverse of a square matrix $\mathbf{M}$ is any matrix such $\mathbf{M}^{-1}$ that $\mathbf{M}\mathbf{M}^{-1} = \mathbf{M}^{-1}\mathbf{M} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix. For a 2 x 2 matrix, if

$$\mathbf{M} = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \quad \Rightarrow \quad \mathbf{M}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

Note that

$$\frac{1}{ad-bc}\begin{bmatrix} a & c \\ b & d \end{bmatrix}\begin{bmatrix} d & -c \\ -b & a \end{bmatrix} = \frac{1}{ad-bc}\begin{bmatrix} d & -c \\ -b & a \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Example 3.1

For the yield data given in the previous section:

(a) Write down the normal equations
(b) Given that

$$\left(\mathbf{X}'\mathbf{X}\right)^{-1} = \begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix}$$

obtain the least-squares estimates.

Solution

(a) The normal equations are

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 160 & 160 & 165 & 165 & 170 & 170 & 175 & 175 \\ 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 160 & 1 \\ 1 & 160 & 2 \\ 1 & 165 & 1 \\ 1 & 165 & 2 \\ 1 & 170 & 1 \\ 1 & 170 & 2 \\ 1 & 175 & 1 \\ 1 & 175 & 2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 160 & 160 & 165 & 165 & 170 & 170 & 175 & 175 \\ 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 77 \\ 79 \\ 82 \\ 83 \\ 85 \\ 88 \\ 90 \\ 93 \end{bmatrix}$$

Therefore, the normal equations are

$$\begin{bmatrix} 8 & 1340 & 12 \\ 1340 & 224700 & 2010 \\ 12 & 2010 & 20 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 677 \\ 113620 \\ 1020 \end{bmatrix}$$

(b) The least squares solutions are

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 8 & 1340 & 12 \\ 1340 & 224700 & 2010 \\ 12 & 2010 & 20 \end{bmatrix}^{-1} \begin{bmatrix} 677 \\ 113620 \\ 1020 \end{bmatrix}$$

$$= \begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix} \begin{bmatrix} 677 \\ 113620 \\ 1020 \end{bmatrix}$$

$$= \begin{bmatrix} -67.825 \\ .89 \\ 2.25 \end{bmatrix}$$

3.3 BIAS AND VARIANCE OF THE LEAST-SQUARES ESTIMATORS IN MULTIPLE LINEAR REGRESSION

From Eq. (3-2), the expectation of the least-squares estimators is

$$\begin{aligned} \mathrm{E}\hat{\boldsymbol{\beta}} &= \mathrm{E}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Y} \\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathrm{E}\mathbf{Y} \\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\left(\mathbf{X}\boldsymbol{\beta}\right) \\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\left(\mathbf{X}'\mathbf{X}\right)\boldsymbol{\beta} \\ &= \boldsymbol{\beta} \end{aligned}$$

Therefore, the least-squares estimators are unbiased.

To be able to define the variance of $\hat{\boldsymbol{\beta}}$, the following should be noted:

(1) <u>The meaning of the variance of a vector r.v.</u>. For example, for the (*m* x 1) vector r.v. $\mathbf{U} = [U_1, U_2, ..., U_m]'$, the variance $\operatorname{var}\mathbf{U}$ is defined by

$$\operatorname{var}\mathbf{U}=\begin{bmatrix}\operatorname{var}U_1 & \operatorname{cov}(U_1,U_2) & \operatorname{cov}(U_1,U_3) & \dots & \operatorname{cov}(U_1,U_m)\\ \operatorname{cov}(U_1,U_2) & \operatorname{var}U_2 & \operatorname{cov}(U_2,U_3) & \dots & \operatorname{cov}(U_2,U_m)\\ \operatorname{cov}(U_1,U_3) & \operatorname{cov}(U_2,U_3) & \operatorname{var}U_3 & \dots & \operatorname{cov}(U_3,U_m)\\ \vdots & \vdots & \vdots & & \vdots\\ \operatorname{cov}(U_1,U_m) & \operatorname{cov}(U_2,U_m) & \operatorname{cov}(U_3,U_m) & \dots & \operatorname{var}U_m\end{bmatrix},$$

where $\operatorname{var}U_i=\mathrm{E}U_i^2-\mathrm{E}^2U_i$ and $\operatorname{cov}(U_i,U_j)=\mathrm{E}U_iU_j-\mathrm{E}U_i\mathrm{E}U_j$. Therefore, $\operatorname{var}\mathbf{U}$ is an *m* x *m* matrix and is known as a variance-covariance matrix. All such matrices are symmetric (i.e. they are equal to their transpose)

For example, suppose $\mathbf{U}=[U_1,U_2]^{/}$, $\operatorname{var}U_1=.125$, $\operatorname{var}U_2=.277$ and $\operatorname{cov}(U_1,U_2)=-.250$, then

$$\operatorname{var}\mathbf{U}=\begin{bmatrix}.125 & -.250\\ -.250 & .277\end{bmatrix}$$

(2) The Variance of the product of a matrix and a vector. If $\mathbf{V}_{(\mathrm{m\,x1})}=\mathbf{C}_{(\mathrm{m\,x\,m})}\mathbf{U}_{(\mathrm{m\,x1})}$, where **C** is a matrix of constants, then

$$\operatorname{var}\mathbf{V}=\operatorname{var}(\mathbf{CU})=\mathbf{C}(\operatorname{var}\mathbf{U})\mathbf{C}^{/}$$

For example, with **U** as above, let

$$\mathbf{C}=\begin{bmatrix}2.12 & 4.35\\ 1.33 & 5.22\end{bmatrix}.$$

Then,

$$\begin{aligned}\operatorname{var}\mathbf{V}&=\mathbf{C}(\operatorname{var}\mathbf{U})\mathbf{C}^{/}\\ &=\begin{bmatrix}2.12 & 4.35\\ 1.33 & 5.22\end{bmatrix}\begin{bmatrix}.125 & -.250\\ -.250 & .277\end{bmatrix}\begin{bmatrix}2.12 & 1.33\\ 4.35 & 5.22\end{bmatrix}\\ &=\begin{bmatrix}1.19 & 2.43\\ 2.43 & 4.30\end{bmatrix}\end{aligned}$$

To calculate the variance of $\hat{\boldsymbol{\beta}}$, we proceed as follows

$$
\begin{aligned}
\operatorname{var}\hat{\boldsymbol{\beta}} &= \operatorname{var}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Y} \\
&= \left\{\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right\}\left(\operatorname{var}\mathbf{Y}\right)\left\{\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right\}' \\
&= \left\{\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right\}\left(\sigma^2\mathbf{I}\right)\left\{\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right\} \\
&= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\left(\mathbf{X}'\mathbf{X}\right)\left(\mathbf{X}'\mathbf{X}\right)^{-1}\sigma^2
\end{aligned}
$$

Therefore,

$$\operatorname{var}\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\sigma^2 \qquad \textbf{3-6}$$

Note that the variances of the various $\hat{\beta}_i$'s can be obtained from the diagonals of the above matrix and the covariance between $\hat{\beta}_i$ and $\hat{\beta}_j$ can be obtained from the $(i+1, j+1)^{\text{th}}$ off-diagonal element.

Note also that the $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ implies that that each $\hat{\beta}_i$ is a specific linear combination of the Y_i's. A question may arise: is there another linear combination of the Y_i's that also leads to unbiased estimators of the β_i's but achieve a <u>smaller variance</u> than in (3-6)? Under the assumptions that $\mathrm{E}\boldsymbol{\varepsilon} = \mathbf{0}$ and $\operatorname{var}\boldsymbol{\varepsilon} = \sigma^2\mathbf{I}$, it can be shown that, of all the possible linear combinations of the Y_i's that lead to unbiased estimators of the β_i's, the least squares are the ones with <u>the smallest possible variance</u>. This is known as the <u>Gauss-Markov theorem.</u>

Finally, given that σ^2 is usually unknown, how can it be estimated? For the case of simple linear regression, Eq. (2-10) gives the answer. For the general linear model, we proceed as follows:

$$
\begin{aligned}
s^2 &= \hat{\sigma}^2 \\
&= \frac{\sum_{i=1}^{n} \hat{\varepsilon}_i^2}{n-(k+1)} \\
&= \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-(k+1)} \\
&= \frac{\left(\mathbf{Y}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)'\left(\mathbf{Y}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)}{n-(k+1)} \\
&= \frac{\left(\mathbf{Y}'-\hat{\boldsymbol{\beta}}'\mathbf{X}'\right)\left(\mathbf{Y}-\mathbf{X}\hat{\boldsymbol{\beta}}\right)}{n-(k+1)} \\
&= \frac{\mathbf{Y}'\mathbf{Y}-\mathbf{Y}'\mathbf{X}\hat{\boldsymbol{\beta}}-\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{Y}+\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}}{n-(k+1)} \\
&= \frac{\mathbf{Y}'\mathbf{Y}-2\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{Y}+\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{Y}}{n-(k+1)}
\end{aligned}
$$

$$
\therefore \quad s^2 = \frac{\mathbf{Y}'\mathbf{Y}-\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{Y}}{n-(k+1)} \tag{3-7}
$$

Example 3.2

For yield data given at the end of Sec. 3.1

(a) Obtain the value of s^2.

(b) For the general linear model given in (3-4), obtain the standard errors of $\hat{\beta}_1$ and $\hat{\beta}_2$, and the covariance between $\hat{\beta}_1$ and $\hat{\beta}_2$.

Solution

(a) From Example 3.1, we have

$$\mathbf{X}'\mathbf{Y} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 160 & 160 & 165 & 165 & 170 & 170 & 175 & 175 \\ 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 77 \\ 79 \\ 82 \\ 83 \\ 85 \\ 88 \\ 90 \\ 93 \end{bmatrix}$$

$$= \begin{bmatrix} 677 \\ 113,620 \\ 1020 \end{bmatrix},$$

and

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} -67.825 & .89 & 2.25 \end{bmatrix}'$$

Moreover,

$$\mathbf{Y}'\mathbf{Y} = \begin{bmatrix} 77 & 79 & 82 & 83 & 85 & 88 & 90 & 93 \end{bmatrix} \begin{bmatrix} 77 \\ 79 \\ 82 \\ 83 \\ 85 \\ 88 \\ 90 \\ 93 \end{bmatrix}$$

$$= 57,501$$

Therefore,

$$s^2 = \frac{\mathbf{Y}'\mathbf{Y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{Y}}{n - (k+1)}$$

$$= \frac{57,501 - \begin{bmatrix} -67.825 & .89 & 2.25 \end{bmatrix} \begin{bmatrix} 677 \\ 113,620 \\ 1020 \end{bmatrix}}{8-3}$$

$$= .345$$

(b) Once σ^2 has been estimated, the estimated $\text{var}(\hat{\boldsymbol{\beta}})$ matrix can be obtained as follows:

$$\text{var }\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{X}\right)^{-1} s^2$$

From Example 3.1,

$$\left(\mathbf{X}'\mathbf{X}\right)^{-1} = \begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix}$$

Therefore,

$$\begin{aligned} \text{var }\hat{\boldsymbol{\beta}} &= \begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix}(.345) \\ &= \begin{bmatrix} 39.151 & -.23115 & -.25875 \\ -.23115 & .00138 & 0 \\ -.25875 & 0 & .1725 \end{bmatrix} \end{aligned}$$

Hence, $\text{SE}\hat{\beta}_1 = \sqrt{.00138} = .0371$, $\text{SE}\hat{\beta}_2 = \sqrt{.1725} = .4153$ and $\text{cov}(\hat{\beta}_1, \hat{\beta}_2) = 0$.

3.4 HYPOTHESIS TESTING IN MULTIPLE LINEAR RGERESSION

Eq. (3-7) enables us to calculate the residual sum of squares in multiple linear regression as

$$SS_{res} = \mathbf{Y}'\mathbf{Y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{Y} \qquad \textbf{3-8}$$

Since the total sum of squares is

$$\begin{aligned} SS_{total} &= S_{YY} \\ &= \sum_i (Y_i - \bar{Y})^2 \\ &= \sum_i Y_i^2 - n\bar{Y}^2 \\ &= \mathbf{Y}'\mathbf{Y} - n\bar{Y}^2, \end{aligned} \qquad \textbf{3-9}$$

we can calculate the regression sum of squares from

$$SS_{reg} = SS_{total} - SS_{res}$$

We are now able to perform an ANOVA (as in Sec. 2.7) and test for the overall significance of regression.

Under normal theory assumptions,

$$\frac{SS_{res}}{\sigma^2} \sim \chi^2_{n-k-1}$$

Under $H_0 : \beta_1 = \beta_2 = ... = \beta_k = 0$ (no overall regression effect),

$$\frac{SS_{reg}}{\sigma^2} \sim \chi^2_k$$

Therefore, under $H_0 : \beta_1 = \beta_2 = ... = \beta_k = 0$,

$$F_{test} = \frac{SS_{reg} / k}{SS_{res} / (n-k-1)} = \frac{MS_{reg}}{s^2} \sim F_{k,n-k-1}.$$

Example 3.3

Perform an ANOVA for the yield data given at the end of Sec. 3.1 and test for the overall significance of regression.

Solution

From Example 3.2, we have

$$\begin{aligned} SS_{res} &= \mathbf{Y}'\mathbf{Y} - \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{Y} \\ &= 57{,}501 - \begin{bmatrix} -67.825 & .89 & 2.25 \end{bmatrix} \begin{bmatrix} 677 \\ 113{,}620 \\ 1020 \end{bmatrix} \\ &= 1.725 \end{aligned}$$

Also,

$$\begin{aligned} SS_{total} &= \mathbf{Y}'\mathbf{Y} - n\overline{Y}^2 \\ &= 57{,}501 - 8(84.625)^2 \\ &= 209.875 \end{aligned}$$

Therefore,

$$\begin{aligned} SS_{reg} &= SS_{total} - SS_{res} \\ &= 209.875 - 1.725 \\ &= 208.15 \end{aligned}$$

Therefore, under $H_0 : \beta_1 = \beta_2 = 0$,

$$F_{test} = \frac{208.15/2}{1.725/5} = 301.667 \sim F_{2,5}$$

From the *F*-tables on pp. 23 and 24, p_{val}=$\Pr\{F_{2,5} > 301.667\} < .01$ so that we reject $H_0 : \beta_1 = \beta_2 = 0$ at the 5% level of significance.

Now that the standard errors of the $\hat{\beta}_i$'s have can be calculated (see Eqs. (3-6) and (3-7), and also Example 3.2), it is straightforward to test hypotheses and build C.I.'s on the β_i's, just as was done in Examples 2.3 and 2.4 for simple linear regression. Under $H_0 : \beta_i = 0$,

$$t_{\text{test}} = \frac{\hat{\beta}_i}{\text{SE}\hat{\beta}_i} \sim t_{n-(k+1)}$$

and a $100(1-\alpha)\%$ C.I. for β_i is

$$\hat{\beta}_i \pm t_{n-(k+1);\alpha/2}\text{SE}\hat{\beta}_i$$

Example 3.4

For the yield data at the end of Sec. 3.1, test the hypothesis that $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$ and obtain the *p*-value of the test. Also, build a 95% CI for β_1.

Solution

From Example 3.1, $\hat{\beta}_1 = .89$ and from Example 3.2, $\mathrm{SE}\hat{\beta}_1 = .0371$. Therefore, under $H_0 : \beta_1 = 0$,

$$t_{test} = \frac{\hat{\beta}_1}{\mathrm{SE}\hat{\beta}_1} = \frac{.89}{.0371} = 23.99 \sim t_5$$

The two-sided p-value is $2\Pr\{t_5 > 23.99\} < .0001$ and $H_0 : \beta_1 = 0$ is rejected at the 5% level.

A 95% CI for β_1 is

$$\begin{aligned}\hat{\beta}_1 \pm t_{5;.025}\mathrm{SE}\hat{\beta}_1 &= \left(.89 - 2.571 \times .0371, .89 + 2.571 \times .0371\right) \\ &= (.795, .985)\end{aligned}$$

Finally, it is also possible to calculate the coefficient of determination in multiple regression. From the sums of squares in Eqs. (3-8) and (3-9),

$$R^2 = \frac{SS_{reg}}{SS_{total}} = 1 - \frac{SS_{res}}{SS_{total}}$$

(see also Eq. (2-20)). As more predictors are added, SS_{reg} increases and R^2 can be made artificially large. This issue is usually addressed by using the adjusted R^2 according to the formula:

$$R_{adj}^2 = 1 - \frac{SS_{res} / \left(n - k - 1\right)}{SS_{total} / \left(n - 1\right)}.$$

Example 3.5

For the yield data at the end of Sec. 3.1, calculate the values of R^2 and R_{adj}^2.

Solution

From Example 3.3, $SS_{total} = 209.875$ and $SS_{reg} = 208.15$, so that

$$R^2 = \frac{SS_{reg}}{SS_{total}} = \frac{208.15}{209.875} = 99.178\% .$$

Thus, 99.178% of the total variation in Y is explained by regression. Moreover,

$$R^2_{adj} = 1 - \frac{SS_{res}/(n-k-1)}{SS_{total}/(n-1)} = 1 - \frac{1.725/(8-3)}{209.875/(8-1)} = 98.844\%$$

3.5 CONFIDENCE AND PREDICTION INTERVALS IN MULTIPLE REGRESSION

Given new values $x_1 = x_{1,0}, x_2 = x_{2,0}, ..., x_k = x_{k,0}$, prediction intervals for a new observation on Y at $\mathbf{x} = \mathbf{x}_0 = \left[1, x_{1,0}, x_{2,0}, ..., x_{k,0}\right]^{/}$ and confidence intervals for the mean value of Y at $\mathbf{x} = \mathbf{x}_0$ can be constructed by using a reasoning similar to that in Sec.2.6. First note that the fitted value at $\mathbf{x} = \mathbf{x}_0$ is

$$\hat{Y}_0 = \mathbf{x}_0^{/}\hat{\boldsymbol{\beta}}$$

The variance of the fitted value $\hat{Y}_0$ is

$$\begin{aligned}
\operatorname{var}\hat{Y}_0 &= \operatorname{var}\mathbf{x}_0^{/}\hat{\boldsymbol{\beta}} \\
&= \mathbf{x}_0^{/}\left(\operatorname{var}\hat{\boldsymbol{\beta}}\right)\mathbf{x}_0 \\
&= \sigma^2\mathbf{x}_0^{/}\left(\mathbf{X}^{/}\mathbf{X}\right)^{-1}\mathbf{x}_0
\end{aligned}$$

Therefore, a 95% prediction interval on the new observation at $\mathbf{x} = \mathbf{x}_0$ is

$$\hat{Y}_0 \pm t_{n-k-1;\alpha/2}\, s\sqrt{1 + \mathbf{x}_0^{/}\left(\mathbf{X}^{/}\mathbf{X}\right)^{-1}\mathbf{x}_0} \qquad \textbf{3-10}$$

and a $100(1-\alpha)\%$ confidence interval on the mean value of Y at $\mathbf{x} = \mathbf{x}_0$ is

$$\hat{Y}_0 \pm t_{n-k-1;\alpha/2}\, s\sqrt{\mathbf{x}_0^{/}\left(\mathbf{X}^{/}\mathbf{X}\right)^{-1}\mathbf{x}_0} \qquad \textbf{3-11}$$

Example 3.6

For the yield data at the end of Sec. 3.1, obtain a 95% prediction interval on a new observation taken at $x_1 = 165$ and $x_2 = 1.5$ and also the corresponding a 95% confidence interval on the mean response. Use the following information:

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} -67.825 \\ .89 \\ 2.25 \end{bmatrix}, \quad \left(\mathbf{X}'\mathbf{X}\right)^{-1} = \begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix}, \quad s^2 = .345$$

Solution

The predicted value of Y at $\mathbf{x}_0 = [1,\ 165,\ 1.5]'$ is

$$\begin{aligned} \hat{Y}_0 &= \mathbf{x}_0'\hat{\boldsymbol{\beta}} \\ &= [1,\ 165,\ 1.5]\begin{bmatrix} -67.825 \\ .89 \\ 2.25 \end{bmatrix} \\ &= 82.4 \end{aligned}$$

A 95% prediction interval on a new observation at $\mathbf{x}_0 = [1,\ 165,\ 1.5]'$ is

$$\begin{aligned} &\hat{Y}_0 \pm t_{n-k-1;\alpha/2}\, s\sqrt{1 + \mathbf{x}_0'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{x}_0} \\ &= \left(82.4 - 2.571\sqrt{.345}\sqrt{1 + [1,\ 165,\ 1.5]\begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix}\begin{bmatrix} 1 \\ 165 \\ 1.5 \end{bmatrix}}, \right. \\ &\qquad \left. 82.4 - 2.571\sqrt{.345}\sqrt{1 + [1,\ 165,\ 1.5]\begin{bmatrix} 113.48 & -.67 & -.75 \\ -.67 & .004 & 0 \\ -.75 & 0 & .5 \end{bmatrix}\begin{bmatrix} 1 \\ 165 \\ 1.5 \end{bmatrix}} \right) \\ &= \left(82.4 - 2.571\sqrt{.345}\sqrt{1 + .155}, 82.4 + 2.571\sqrt{.345}\sqrt{1 + .155}\right) \\ &= (80.78, 84.02) \end{aligned}$$

The corresponding 95% confidence interval in the mean response at $\mathbf{x}_0 = [1,\ 165,\ 1.5]'$ is

$$\hat{Y}_0 \pm t_{n-k-1;\alpha/2} s \sqrt{\mathbf{x}_0^{/} \left(\mathbf{X}^{/}\mathbf{X}\right)^{-1} \mathbf{x}_0}$$
$$= \left(82.4 - 2.571\sqrt{.345}\sqrt{.155}, 82.4 + 2.571\sqrt{.345}\sqrt{.155}\right)$$
$$= \left(81.81, 82.99\right)$$

4 ADDITIONAL TOPICS IN MULTIPLE LINEAR REGRESSION

4.1 THE ISSUE OF MULTICOLLINEARITY

Consider the following data:

Y	X1	X2
5.15	10.00	20.01
6.30	11.00	22.00
8.30	11.90	23.80
9.90	12.70	25.40
10.30	13.30	26.60
11.10	14.20	28.40
12.30	14.70	29.40
14.00	15.00	30.00

Note that $X_2 \approx 2X_1$ so that X_2 works as a proxy for X_1 ($r_{12} \approx 1$). Thus, instead of fitting the multiple linear regression model $\mathrm{E}(Y \mid X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$, it is wiser to fit the simple linear regression $\mathrm{E}(Y \mid X_1) = \beta_0 + \beta_1 X_1$. If this is done, the following is obtained:

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval Lower	Upper	Hypothesis Test df	Sig.
(Intercept)	-11.295	1.3065	-13.856	-8.734	1	<.001
X1	1.631	.1008	1.434	1.829	1	<.001

Dependent Variable: Y
Model: (Intercept), X1

This shows that there is a statistically significant relationship between *Y* and *X*; each unit increase in X_1 is associated with a 1.631 unit increase in the mean value of *Y*.

Suppose the investigator did not pay attention to the fact that $X_2 \approx 2X_1$ and decided to fit the multiple linear regression model $\mathrm{E}(Y \mid X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$. If this is done, the following is obtained:

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval Lower	Upper	Hypothesis Test df	Sig.
(Intercept)	-11.740	1.7362	-15.143	-8.337	1	<.001
X1	-49.181	132.1421	-308.175	209.812	1	.710
X2	25.422	66.1128	-104.156	155.001	1	.701

Dependent Variable: Y
Model: (Intercept), X1, X2

Note the huge standard errors of $\hat{\beta}_1$ and $\hat{\beta}_2$, resulting in both X_1 and X_2 losing significance. The least-squares estimates are no longer reliable.

Yet another problem arises when the response is modified only slightly. Suppose the first response is changed from 5.15 to 5.10, as follows:

Y	X1	X2
5.10	10.00	20.01
6.30	11.00	22.00
8.30	11.90	23.80
9.90	12.70	25.40
10.30	13.30	26.60
11.10	14.20	28.40
12.30	14.70	29.40
14.00	15.00	30.00

If the multiple linear regression model $\mathrm{E}(Y \mid X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ is run again, the following is obtained:

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval Lower	Upper	Hypothesis Test df	Sig.
(Intercept)	-11.740	1.7362	-15.143	-8.337	1	<.001
X1	-39.181	132.1421	-298.175	219.812	1	.767
X2	20.422	66.1128	-109.156	150.001	1	.757

Dependent Variable: Y
Model: (Intercept), X1, X2

Note the big changes in the values $\hat{\beta}_1$ and $\hat{\beta}_2$. This means that the least-squares estimates are highly unstable.

In general, when one predictor is a linear combination of other predictors in multiple linear regression, the least squares estimators of the regression coefficients become unreliable and unstable. This is known as multicollinearity.

It might seem at first that running a simple bivariate correlation matrix between the predictors in multiple linear regression would be sufficient to detect multicollinearity. We could simply look for high correlations and remove one of the two highly correlated variables. However, this strategy is deficient because multicollinearity occurs not only when two predictors are highly correlated (i.e. one is almost a linear function of the other) but also when one predictor is a linear function of several other predictors. For this reason, an alternative measure is required and the one that is most frequently used is called the variance inflation factor (VIF), defined for the *j*th regression coefficient as

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination when X_j is regressed on all other predictors in the multiple regression model. A large R_j^2 (resulting in a large VIF_j) indicates increased likelihood of multicollinearity in the model, and a potentially unreliable and unstable least squares estimate of β_j. The cut-off used for VIF_j is usually 10, i.e. $\mathrm{VIF}_j > 10$ is taken as an indication of multicollinearity.

Once problematic VIF_j's have been identified, there are two commonly used remedial solutions: (1) Dropping the problematic predictor, (ii) Using ridge regression on all predictors. We will illustrate the first example though an example.

Example 4.1

Consider the following data set. Using a software, fit a multiple linear regression and find the VIF for each regression coefficient. Devise a strategy of dropping predictors until a stable model is achieved.

y	x1	x2	x3	x4
5.195	1.029	.752	1.157	.938
-.973	-.220	-.335	-.429	-.516
4.385	1.404	1.231	1.491	.096
8.437	3.151	2.899	3.402	-.462
-2.121	-.606	-.499	-.430	-.434
-1.953	-.562	-.370	-.210	-.309
10.411	3.210	2.872	3.200	.222
4.358	1.586	1.441	1.425	-.479
-.254	-.887	-.878	-1.253	1.256
2.802	1.470	1.093	1.364	-.895
-2.231	-.870	-.795	-.875	-.187
-3.015	-.818	-1.020	-.986	-.440
4.946	.579	.276	.550	1.447
-11.334	-3.761	-3.482	-4.187	.197
-8.028	-3.481	-3.237	-3.255	1.032
-5.275	-1.049	-1.158	-1.161	-1.486
-6.234	-2.103	-1.738	-2.111	.267
2.907	.605	.611	.769	.890
-5.631	-1.865	-1.584	-1.835	.082
-6.639	-2.816	-2.675	-2.932	1.065

Solution

Using IBM SPSS (vers 29), the following model was fitted by using Analyze →Regression → Linear, entering the variables and clicking on both Estimates and Collinearity diagnostics under Statistics.

$$\mathrm{E}\left(Y \mid \mathbf{X}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	t	Sig.	Collinearity Statistics VIF
1	(Constant)	.113	.095	1.184	.255	
	x1	3.807	.628	6.066	<.001	204.453
	x2	-1.511	.745	-2.027	.061	236.569
	x3	.562	.507	1.108	.285	142.259
	x4	1.723	.113	15.290	<.001	1.056

a. Dependent Variable: y

It is seen that there is serious multicollinearity for x_1, x_2, x_3. We drop x_2 (the variable with the highest VIF) and re-run the model. The following was obtained:

Coefficients[a]

Model		Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics VIF
1	(Constant)	.197	.093		2.105	.051	
	x1	2.945	.504	.995	5.839	<.001	110.574
	x3	.074	.488	.026	.151	.881	110.171
	x4	1.735	.123	.235	14.102	<.001	1.054

a. Dependent Variable: y

There are still problems with x_1 and x_3. We drop x_1 and re-run the model:

Coefficients[a]

Model		Unstandardized Coefficients B	Std. Error	Standardized Coefficients Beta	t	Sig.	Collinearity Statistics VIF
1	(Constant)	.215	.160		1.338	.198	
	x3	2.908	.081	1.014	35.698	<.001	1.043
	x4	1.663	.210	.225	7.915	<.001	1.043

a. Dependent Variable: y

The model is stable now, with only x_3 and x_4 as relevant. Both turn out to be significant.

4.2 THE USE OF DUMMY VARIABLES FOR CATEGORICAL VARIABLES

So far, the predictors used have been *quantitative* (i.e., measured on *a continuous* scale), such as age, height, weight, temperature, etc. However, many variables of interest are also *categorical* or *qualitative* such as age group, geographical region, cellphone type, eye color, etc.

<u>Categorical Variables with Two Categories (binary/dichotomous variables)</u> This most common situation is a categorical variable with two categories (or levels). For example, in modeling the HbA1C of a subject as a function of the average number of hours of exercise per day, we might also be interested in including sex as an additional predictor. The presence or absence of a given factor might also be used as a categorical variable. In the example above, we might also be interested in adding the diabetic status of the subject in our model. When regression is performed on categorical variable (s) while controlling (or adjusting) for a continuous variable, the analysis is known as analysis of covariance (ANCOVA).

Consider a model with outcome Y_i, continuous predictor (or *covariate*) X_i and binary variable Z_i defined by

$$Z_i = \begin{cases} 1 & \text{if } i\text{th observation is in first category} \\ 0 & \text{if } i\text{th observation is in second category} \end{cases}$$

The regression model is written as

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \varepsilon_i, \quad i = 1, 2, ..., n \qquad \textbf{4-1}$$

The **X** matrix is written as

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & 1 \\ 1 & x_2 & 1 \\ 1 & x_3 & 1 \\ \vdots & \vdots & \vdots \\ 1 & & 1 \\ 1 & & 0 \\ 1 & & 0 \\ 1 & & 0 \\ \vdots & \vdots & \vdots \\ 1 & x_n & 0 \end{bmatrix} \begin{array}{l} \left.\begin{array}{l} \\ \\ \\ \\ \\ \end{array}\right\}\text{category 1} \\ \left.\begin{array}{l} \\ \\ \\ \\ \\ \end{array}\right\}\text{category 2} \end{array}$$

Once the **X** matrix is obtained, using the **Y** vector all previous calculations can be repeated, e.g. $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, etc.

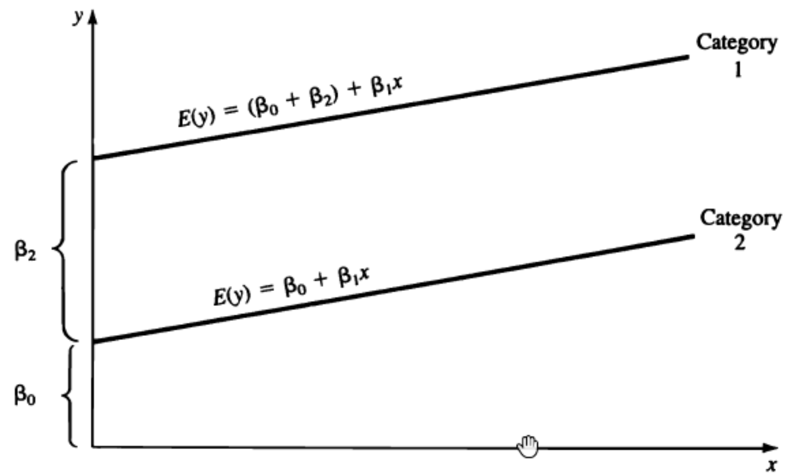
It is important to note that the model in (4-1) embodies two linear models, one for category 1 and the other for category 2, as follows:

For category 1:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 (1) + \varepsilon = (\beta_0 + \beta_2) + \beta_1 x_i + \varepsilon, \quad i = 1, 2, ..., n$$

For category 2:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 (0) + \varepsilon_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, 2, ..., n$$



It is seen that two lines are parallel with the same slope β_1. Therefore, the model in (4-1) implicitly assumes that the mean change in *Y* for each unit change in *x* is the same for both categories. This assumption will soon be relaxed by introducing interactions in the model.

The two parallel lines have an intercept shift of β_2. If the effect of category is non-significant, then $\beta_2 = 0$.

Example 4.2

An experiment was conducted on the impact of grazing on the seed production of a biennial plant. Forty plants were allocated to two treatments, grazed (by rabbits) and ungrazed. The response was fruit production (dry weight in mg) (*Y*) for each of the 40 plants and one covariate (*X*) used was the diameter of the top of the rootstock.

	Diameter	Production	Grazing
1	6.225	59.77	Ungrazed
2	6.487	60.98	Ungrazed
3	4.919	14.73	Ungrazed
4	5.130	19.28	Ungrazed
5	5.417	34.25	Ungrazed
6	5.359	35.53	Ungrazed
7	7.614	87.73	Ungrazed
8	6.352	63.21	Ungrazed
9	4.975	24.25	Ungrazed
10	6.930	64.34	Ungrazed
11	6.248	52.92	Ungrazed
12	5.451	32.35	Ungrazed
13	6.013	53.61	Ungrazed
14	5.928	54.86	Ungrazed
15	6.264	64.81	Ungrazed
16	7.181	73.24	Ungrazed
17	7.001	80.64	Ungrazed
18	4.426	18.89	Ungrazed
19	7.302	75.49	Ungrazed
20	5.836	46.73	Ungrazed
21	10.253	116.05	Grazed
22	6.958	38.94	Grazed
23	8.001	60.77	Grazed
24	9.039	84.37	Grazed
25	8.910	70.11	Grazed
26	6.106	14.95	Grazed
27	7.691	70.70	Grazed
28	8.988	80.31	Grazed
29	8.975	82.35	Grazed
30	9.844	105.07	Grazed
31	8.508	73.79	Grazed
32	7.354	50.08	Grazed
33	8.643	78.28	Grazed
34	7.916	41.48	Grazed
35	9.351	98.47	Grazed
36	7.066	40.15	Grazed
37	8.158	52.26	Grazed
38	7.382	46.64	Grazed
39	8.515	71.01	Grazed
40	8.530	83.03	Grazed

(a) By defining a suitable categorical variable for 'Grazing,' write down the first and last three rows of the **X** matrix for the data above.

(b) By using a statistical software, obtain the ANOVA table and line of best fit for the model

$$\mathrm{E}(Y \mid \mathbf{X}) = \beta_0 + \beta_1 X + \beta_2 Z$$

(c) Write down the separate models for grazed and ungrazed plants

(d) Is the effect of grazing significant at the 5% level? Interpret your finding.

Solution

(a) Let

$$Z_i = \begin{cases} 1 & \text{if } i\text{th plant is grazed} \\ 0 & \text{if } i\text{th plant is ungrazed} \end{cases}$$

Then,

$$\mathbf{X} = \begin{bmatrix} 1 & 6.225 & 0 \\ 1 & 6.487 & 0 \\ 1 & 4.919 & 0 \\ \vdots & \vdots & \vdots \\ 1 & 7.382 & 1 \\ 1 & 8.515 & 1 \\ 1 & 8.530 & 1 \end{bmatrix}$$

(b) IBM Statistics 29.0.0.0 gives the following ANOVA table:

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	22059.375	2	11029.688	242.272	<.001[b]
	Residual	1684.461	37	45.526		
	Total	23743.837	39			

a. Dependent Variable: Production

b. Predictors: (Constant), Grazing, Diameter

and least-squares estimates:

Model		Unstandardized Coefficients		t	Sig.	Collinearity Statistics
		B	Std. Error			VIF
1	(Constant)	-91.726	7.115	-12.892	<.001	
	Diameter	23.560	1.149	20.509	<.001	2.476
	Z	-36.103	3.357	-10.753	<.001	2.476

a. Dependent Variable: Production

The line-of-best-fit is

$$\hat{Y} = -91.726 + 23.560X - 36.103Z$$

(c) For grazed plants, $Z = 1$,

$$\begin{aligned}\hat{Y} &= -91.726 + 23.560X - 36.103(1) \\ &= -127.829 + 23.560X\end{aligned}$$

For ungrazed plants, $Z = 0$,

$$\begin{aligned}\hat{Y} &= -91.726 + 23.560X - 36.103(0) \\ &= -91.726 + 23.560X\end{aligned}$$

(d) Since pval < .001, the effect of grazing is significant at the 5% level. For any given value of X, the mean fruit production for grazed plants was lower than for ungrazed plants by 127.829-91.726 = 36.1.

<u>Categorical Variables with More Than Two Categories</u>

Suppose in Example 4.1, the categorical variable 'Grazing' was made up of <u>three</u> categories, namely Ungrazed, Moderately Grazed, and Fully Grazed. We then proceed by defining the following <u>two dummy variables (or indicators)</u> Z_1 and Z_2 as follows:

$$Z_1 = \begin{cases} 1 & \text{if fully grazed} \\ 0 & \text{otherwise} \end{cases}$$

and

$$Z_2 = \begin{cases} 1 & \text{if moderately grazed} \\ 0 & \text{otherwise} \end{cases}$$

Then the three categories correspond to:

- fully grazed: $Z_1 = 1, Z_2 = 0$
- moderately grazed: $Z_1 = 0, Z_2 = 1$
- ungrazed: $Z_1 = 0, Z_2 = 0$

The model in (4-1) then becomes

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_{1i} + \beta_3 Z_{2i} + \varepsilon_i, \qquad i = 1, 2, ..., n$$

and the **X** matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & 1 & 0 \\ 1 & x_2 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & & 1 & 0 \\ 1 & & 0 & 1 \\ 1 & & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & & 0 & 1 \\ 1 & & 0 & 0 \\ 1 & & 0 & 0 \\ 1 & x_{n-1} & \vdots & \vdots \\ 1 & x_n & 0 & 0 \end{bmatrix} \begin{array}{l} \left.\vphantom{\begin{matrix}1\\1\\\vdots\\1\end{matrix}}\right\} \text{fully grazed} \\ \left.\vphantom{\begin{matrix}1\\1\\\vdots\\1\end{matrix}}\right\} \text{moderately grazed} \\ \left.\vphantom{\begin{matrix}1\\1\\\vdots\\1\end{matrix}}\right\} \text{ungrazed} \end{array}$$

Example 4.2

Consider the data in Example 4.1 on the effect of grazing on fruit production. Suppose *Y* and *X* remain the same but the categorical variable ‘Grazing’ now has three levels, as follows:

	Diameter	Production	Grazing
1	6.225	59.77	Ungrazed
2	6.487	60.98	Ungrazed
3	4.919	14.73	Ungrazed
4	5.130	19.28	Ungrazed
5	5.417	34.25	Ungrazed
6	5.359	35.53	Ungrazed
7	7.614	87.73	Ungrazed
8	6.352	63.21	Ungrazed
9	4.975	24.25	Ungrazed
10	6.930	64.34	Ungrazed
11	6.248	52.92	Ungrazed
12	5.451	32.35	Ungrazed
13	6.013	53.61	Ungrazed
14	5.928	54.86	Ungrazed
15	6.264	64.81	Ungrazed
16	7.181	73.24	Moderately Grazed
17	7.001	80.64	Moderately Grazed
18	4.426	18.89	Moderately Grazed
19	7.302	75.49	Moderately Grazed
20	5.836	46.73	Moderately Grazed
21	10.253	116.05	Moderately Grazed
22	6.958	38.94	Moderately Grazed
23	8.001	60.77	Moderately Grazed
24	9.039	84.37	Moderately Grazed
25	8.910	70.11	Moderately Grazed
26	6.106	14.95	Moderately Grazed
27	7.691	70.70	Moderately Grazed
28	8.988	80.31	Moderately Grazed
29	8.975	82.35	Moderately Grazed
30	9.844	105.07	Grazed
31	8.508	73.79	Grazed
32	7.354	50.08	Grazed
33	8.643	78.28	Grazed
34	7.916	41.48	Grazed
35	9.351	98.47	Grazed
36	7.066	40.15	Grazed
37	8.158	52.26	Grazed
38	7.382	46.64	Grazed
39	8.515	71.01	Grazed
40	8.530	83.03	Grazed

(a) By defining suitable dummy variables for ‘Grazing’, write down the 1st and 2nd, 15th and 16th , 29th and 30th, and 39th and 40th rows of the **X** matrix for the data above.
(b) By using a statistical software, obtain the ANOVA table and line of best fit for the multiple linear regression model
(c) Write down the separate models for grazed, moderately grazed and ungrazed plants
(d) Test for the various categories of ‘Grazing’ at the 5% level? Interpret your findings.

Solution

(a) Let

$$Z_1 = \begin{cases} 1 & \text{if fully grazed} \\ 0 & \text{otherwise} \end{cases}$$

and

$$Z_2 = \begin{cases} 1 & \text{if moderately grazed} \\ 0 & \text{otherwise} \end{cases}$$

The **X** matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & 6.225 & 0 & 0 \\ 1 & 6.487 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 6.264 & 0 & 0 \\ 1 & 7.181 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 8.988 & 0 & 1 \\ 1 & 9.975 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 8.515 & 1 & 0 \\ 1 & 8.530 & 1 & 0 \end{bmatrix}$$

(b) IBM Statistics 29.0.0.0 gives the following ANOVA table:

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	18915.545	3	6305.182	47.012	<.001[b]
	Residual	4828.292	36	134.119		
	Total	23743.837	39			

and regression output:

Model		Unstandardized Coefficients B	Std. Error	t	Sig.	Collinearity Statistics VIF
1	(Constant)	-61.845	10.550	-5.862	<.001	
	Diameter	18.478	1.699	10.874	<.001	1.839
	Z1	-24.170	6.081	-3.974	<.001	2.199
	Z2	-13.686	5.150	-2.657	.012	1.800

a. Dependent Variable: Production

(c) The line-of-best fit is

$$\hat{Y} = -61.845 + 18.478X - 24.170Z_1 - 13.686Z_2$$

For grazed plants, $Z_1 = 1, Z_2 = 0$, so that

$$\begin{aligned}\hat{Y} &= -61.845 + 18.478X - 24.170(1) - 13.686(0) \\ &= -86.015 + 18.478X\end{aligned}$$

For moderately grazed plants, $Z_1 = 0, Z_2 = 1$, so that

$$\begin{aligned}\hat{Y} &= -61.845 + 18.478X - 24.170(0) - 13.686(1) \\ &= -75.531 + 18.478X\end{aligned}$$

For ungrazed plants, $Z_1 = 0, Z_2 = 0$, so that

$$\begin{aligned}\hat{Y} &= -61.845 + 18.478X - 24.170(0) - 13.686(0) \\ &= -61.845 + 18.478X\end{aligned}$$

(e) The table of coefficients shows that both Z_1 (pval < .001) and Z_2 (pval= .012) are statistically significant at the 5% level. In particular, compared to ungrazed plants, moderately grazed pants have a mean production less by 13.66 on average, and grazed plants have a mean production less by 24.170 on average.

In general, within reasonable limits, a model can have any number of covariates and any number of categorical variables, each with its own number of levels. For example, a model with one covariate X, and two categorical variables, one with four levels and the other with three levels, is modeled as follows:

$$Y = \beta_0 + \beta_1 X + \beta_2 Z_1 + \beta_3 Z_2 + \beta_4 Z_3 + \beta_5 W_1 + \beta_6 W_2 + \varepsilon$$

where (Z_1, Z_2, Z_3) are the dummy variables for the first categorical variable, and (W_1, W_2) are the dummy variables for the second categorical variable.

Doing Away with the Parallel Line Assumption: Testing for Interactions

Recall that, so far, in our modeling with categorical predictors, our models were such that they forced a similar slope for each line within a category. In Example 4.1, for both grazed and ungrazed, we assumed a similar slope and the output showed that, for each category, the increase in mean production for each unit increase in diameter is 23.560. Similarly, in Example 4.2 a similar slope was assumed for all three categories.

Of course, there is no reason why the rate of increase in mean production should be the same for both (or all three) categories. This assumption can easily be removed by allowing for covariate-categorical interactions in the model. For example, the model in (4-1) can be modified to:

$$Y = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i + \varepsilon_i, \quad i = 1, 2, ..., n$$

The **X** matrix for this model is

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 & 1 & x_1 \\ 1 & x_2 & 1 & x_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_i & 1 & x_i \\ 1 & x_{i+1} & 0 & 0 \\ 1 & x_{i+2} & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & 0 & 0 \end{bmatrix}$$

For category 1 (i.e. $Z_i = 1$),

$$\begin{aligned} Y &= \beta_0 + \beta_1 X_i + \beta_2 (1) + \beta_3 X_i (1) + \varepsilon_i \\ &= (\beta_0 + \beta_2) + (\beta_1 + \beta_3) X_i + \varepsilon_i \end{aligned}$$

For category 2 (i.e. $Z_i = 0$),

$$\begin{aligned} Y &= \beta_0 + \beta_1 X_i + \beta_2 (0) + \beta_3 X_i (0) + \varepsilon_i \\ &= \beta_0 + \beta_1 X_i + \varepsilon_i \end{aligned}$$

A test of similar intercepts is a test of $H_0 : \beta_2 = 0$. More importantly, a test for similar slopes across the two categories is a test of $H_0 : \beta_3 = 0$ (no interaction). When there is no interaction, then the change in the mean value of *Y* for each unit increase in *X* is the same when $Z_i = 0$ or when $Z_i = 1$.

Example 4.3

For the data in Example 4.2, use a statistical software to fit a linear regression model that allows for a different slope for each category (i.e. that allows for an *X**Z* interaction). Perform a test of no interaction at the 5% level.

Solution

The model is

$$\mathrm{E}\left(Y \mid \mathbf{X}\right) = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$$

The ANOVA table is

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	22064.187	3	7354.729	157.634	<.001[b]
	Residual	1679.649	36	46.657		
	Total	23743.837	39			

The table of beta coefficients is

Model		Unstandardized Coefficients B	Std. Error	t	Sig.
1	(Constant)	-94.367	10.933	-8.632	<.001
	Diameter	23.996	1.788	13.417	<.001
	Z	-30.806	16.842	-1.829	.076
	Diameter_Z	-.756	2.354	-.321	.750

a. Dependent Variable: Production

The least-squares line is

$$\hat{Y} = -94.367 + 23.996X - 30.806Z - .756XZ$$

A test of equality of slopes is a test of whether the coefficients of *XZ* is zero, i.e. $H_0 : \beta_3 = 0$. Since pval > .05, we cannot reject H_0 at the 5% level, i.e. there is no evidence of an interaction between diameter and grazing at the 5% level.

Important Notes

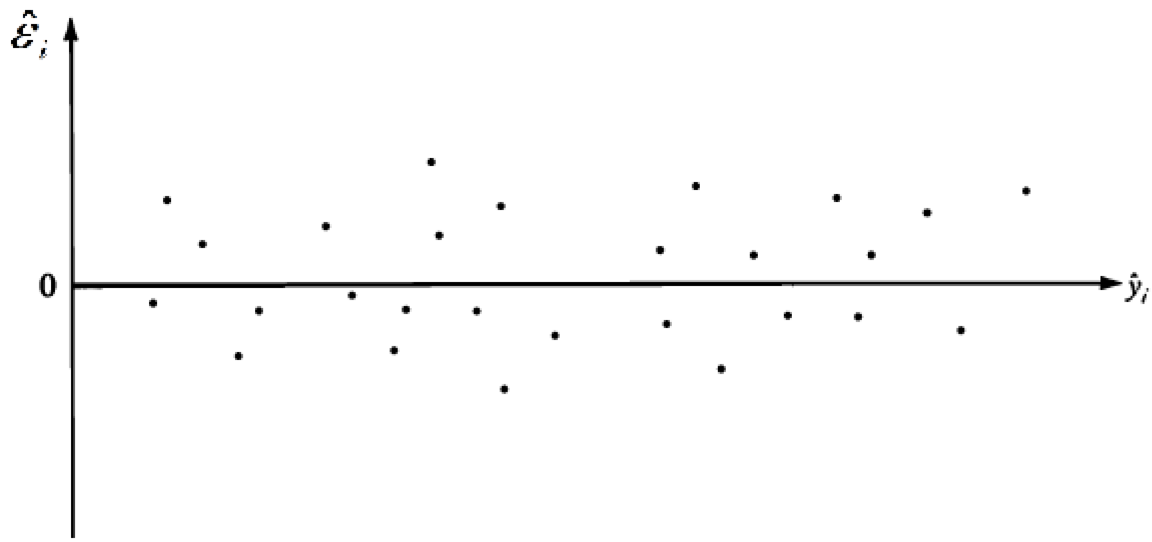
(a) When a model with interactions is fitted, the table of coefficients should be read bottom-up, i.e. the significance of the interactions should be investigated first.
(b) If an *X*Z* interaction is significant, then
    (i) both *X* and *Z* should be retained in the model, irrespective of their *p*-values.
    (ii) Interpreting the effects of *X* and *Z* makes no sense, because the effect of *X* depends on the level of *Z* and the effect of *Z* depends on the value of *X.*
(c) If an *X*Z* interaction is not significant, then the individual p-values of *X* and *Z* may be looked at and their effects interpreted.

Thus, in the table of coefficients for Example 4.3, since the *X*Z* interaction is not significant, we can move to the main effects. Grazing (*Z*) has no significant effect (pval=.076) but diameter (*X*) is significant (pval<.001) – for each unit increase in diameter the mean value of production increases by 23.996.

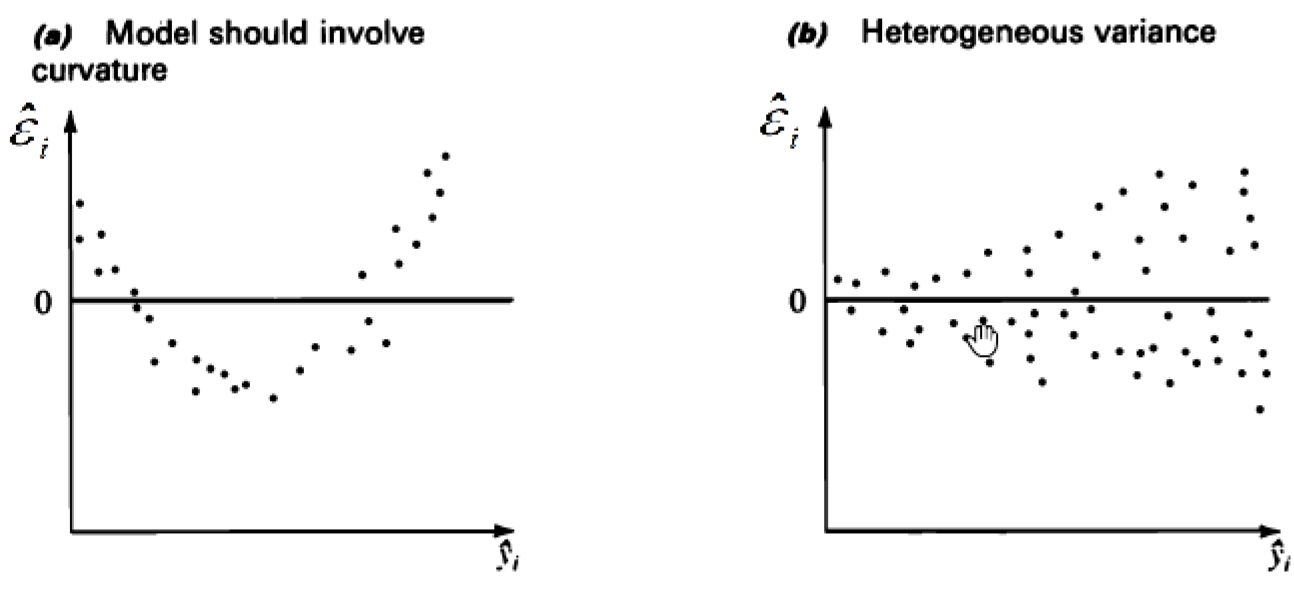
On the other hand, *if* the *X*Z* interaction had been significant (pval < .05) while all other p-values remained the same, then both *X* and *Z* would be retained in the model. For *Z* = 0, the rate of increase in mean production with diameter would be 23.996 and for *Z* = 1, the rate of increase in mean production with diameter would be 23.996 - .756 = 23.243.

4.3 RESIDUAL PLOTS IN LINEAR REGRESSION

If the correct model is fitted and the Gauss-Markov conditions are satisfied, then a plot of the residuals against the fitted values should resemble a null plot, i.e. a random scatter around the x-axis, as shown in the figure below.



Two common problematic situations arise when the wrong model is fitted (figure (a) below) or when the variance is heterogenous (figure (b) below).



(a) <u>Deciding if the wrong model has been fitted: Tukey's test for non-additivity</u>

If the residual plot shows signs of curvature, the significance of the curvature may be tested by performing a <u>Tukey's test for non-additivity.</u> This test proceeds as follows: the original regression, say $\mathrm{E}(Y \mid \mathbf{X}) = \beta_0 + \beta_1 X_1 + ... + \beta_k X_k$ is run and the fitted values $\hat{Y}_i$ stored. A new multiple linear regression is now fitted using $\mathrm{E}(Y \mid \mathbf{X}) = \beta_0 + \beta_1 X_1 + ... + \beta_k X_k + \gamma \hat{Y}_i^2$. If β_{k+1} is significantly different from zero, then additivity is rejected and we conclude that the model shows evidence of curvature.

Example 4.7

Consider the following data:

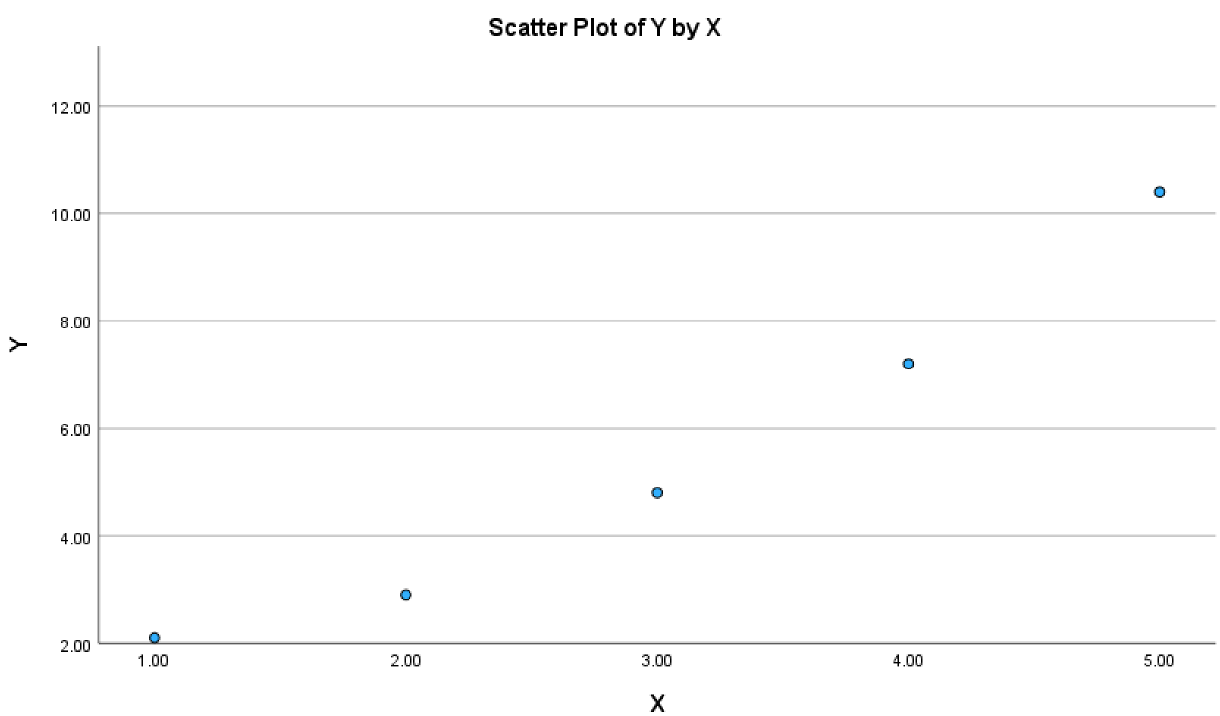
X	1	2	3	4	5
Y	2.1	2.9	4.8	7.2	10.4

Using a statistical software:

(a) do a scatterplot of *Y* against *X*;
(b) fit a straight line, do a residual plot and comment on it;
(c) perform a Tukey's test of non-additivity.
(d) based on your answer in (c), suggest and implement a remedial measure.

Solution

(a) Using IBM SPSS 29.0.0.0,

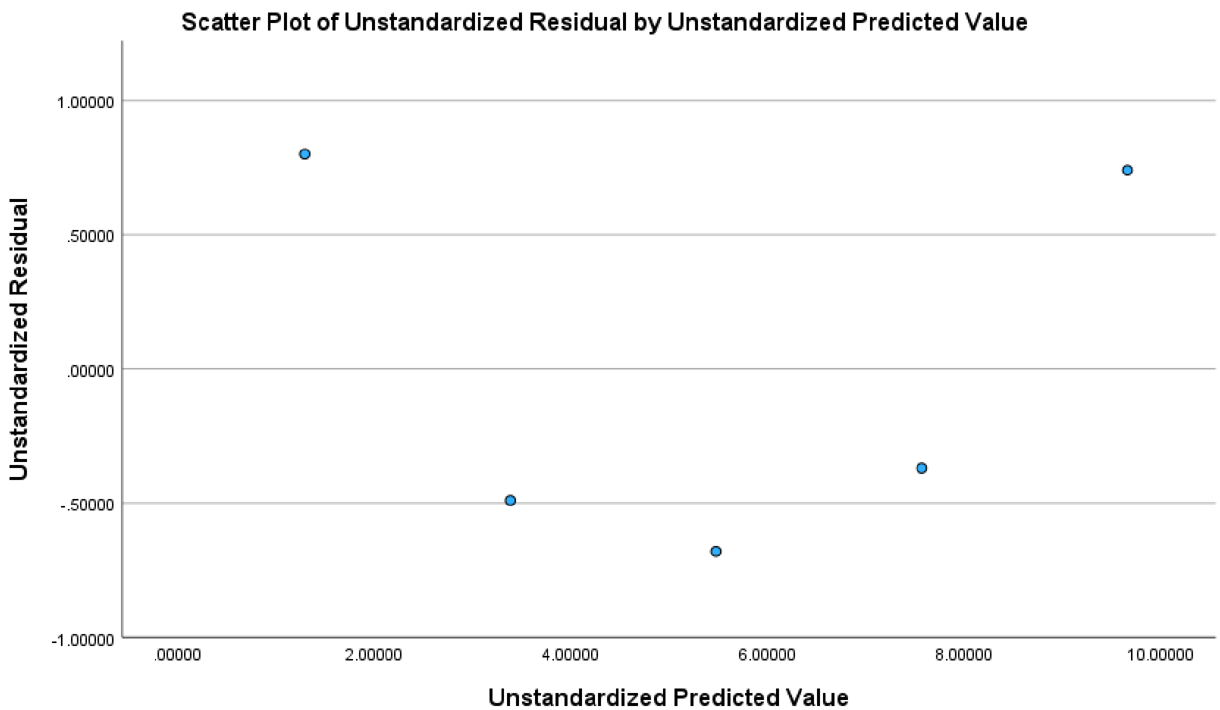


(b) The line-of-best fit is $\hat{Y} = -.790 + 2.090X$:

Model		Unstandardized Coefficients B	Std. Error	t	Sig.
1	(Constant)	-.790	.862	-.916	.427
	X	2.090	.260	8.040	.004

a. Dependent Variable: Y

The residual plot is:



Although the number of data points is limited, there seems to be a curvilinear trend in the residual plot.

(c) The values of *X*, *Y*, $\hat{Y}$ and $\hat{Y}^2$ are tabulated below:

X	Y	Y_hat	Y_hat_2
1.00	2.10	1.30000	1.69
2.00	2.90	3.39000	11.49
3.00	4.80	5.48000	30.03
4.00	7.20	7.57000	57.30
5.00	10.40	9.66000	93.32

A multiple linear regression of $Y = \beta_0 + \beta_1 X + \gamma \hat{Y}^2 + \varepsilon$ is now run, resulting in the following table of beta coefficients:

Model		Unstandardized Coefficients B	Std. Error	t	Sig.
1	(Constant)	1.806	.214	8.433	.014
	X	.105	.146	.719	.547
	Y_hat_2	.087	.006	13.967	.005

a. Dependent Variable: Y

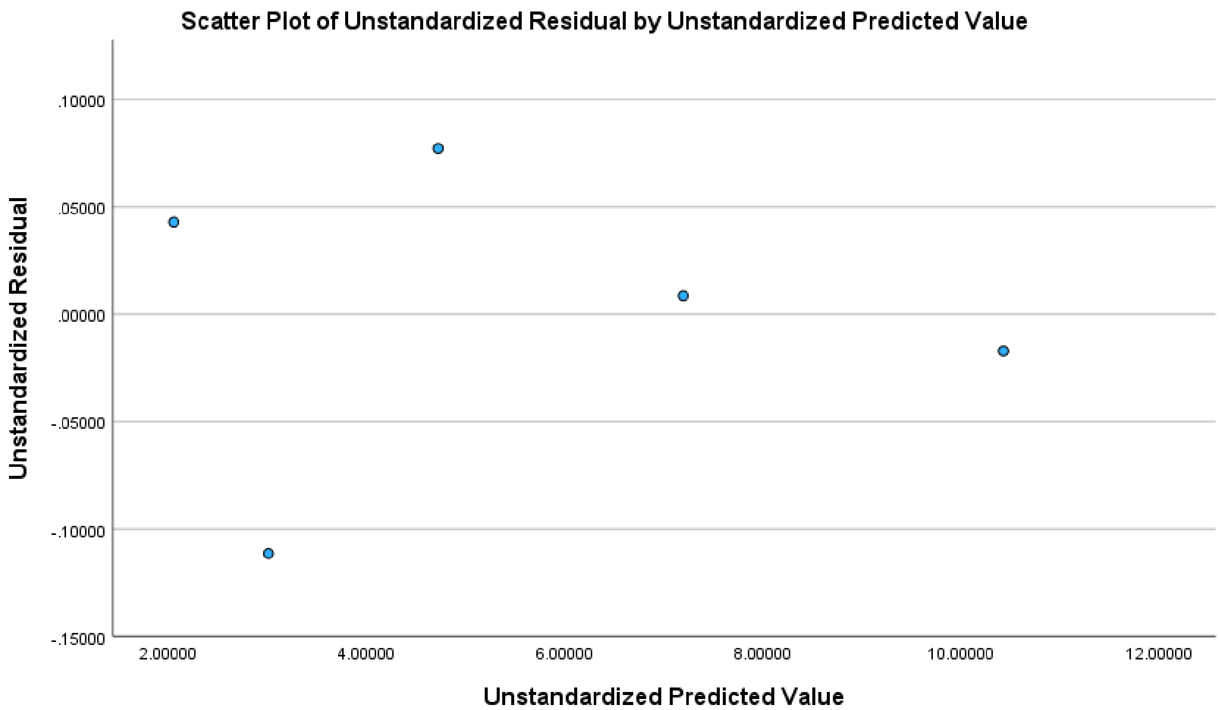
A test of $H_0 : \gamma = 0$ (and therefore additivity) is rejected at the 5% level (pval < .05) and we conclude there is evidence of curvature. A straight line for *Y* against *X* therefore does not seem to appropriate.

(d) Based on the conclusion in (c), instead of $Y = \beta_0 + \beta_1 X + \varepsilon$ we fit the curve $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$. The least-squares curve is $\hat{Y} = 1.860 - .181X + .379X^2$:

Model		Unstandardized Coefficients B	Std. Error	t	Sig.
1	(Constant)	1.860	.218	8.551	.013
	X	-.181	.166	-1.095	.388
	X2	.379	.027	13.967	.005

a. Dependent Variable: Y

The residual plot now looks as follows:



The above is more or less a random scatter, suggesting a quadratic curve is more appropriate for the given data.

In the previous example, we considered the case of simple linear regression: when Tukey's test suggested curvature, we did not have to choose which predictor to square (or cube) since there was only one. The question that automatically arises therefore is the following, which predictor to square when we have more than one predictor? The question may be answered by doing a <u>partial residual plot</u> (or component-plus-residual plot).

For a model $\mathrm{E}(Y \mid \mathbf{X}) = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$, the quantity $PR_j = \hat{\varepsilon} + \hat{\beta}_j X_j$ (where $\hat{\varepsilon}$ is the residual for the given model) is called the partial residual for the predictor X_j and a plot of PR_j vs. X_j is called a partial residual plot for the predictor X_j. (Partial here means that the effects of all *other* predictors have been removed or partialed out). If the plot PR_j vs. X_j is a straight line, then no square (or cube) term is needed for X_j. On the other hand, plot PR_j vs. X_j is a curved, then quadratic or cubic terms for X_j will be needed.

Example 4.8

Consider the following data:

X1	X2	Y
1.00	2.00	5.30
2.00	3.00	7.10
3.00	3.00	8.40
4.00	4.00	10.20
5.00	5.00	12.70
6.00	5.00	15.80
7.00	6.00	19.00
8.00	6.00	22.50
9.00	7.00	26.10
10.00	8.00	31.50

Using a statistical software:

(a) fit a linear model $\mathrm{E}(Y \mid X_1, X_2) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
(b) do a residual plot and comment on it;
(c) perform a Tukey's test of non-additivity;
(d) use partial residual plots to fit a more appropriate model than the one in (a) above.

Solution

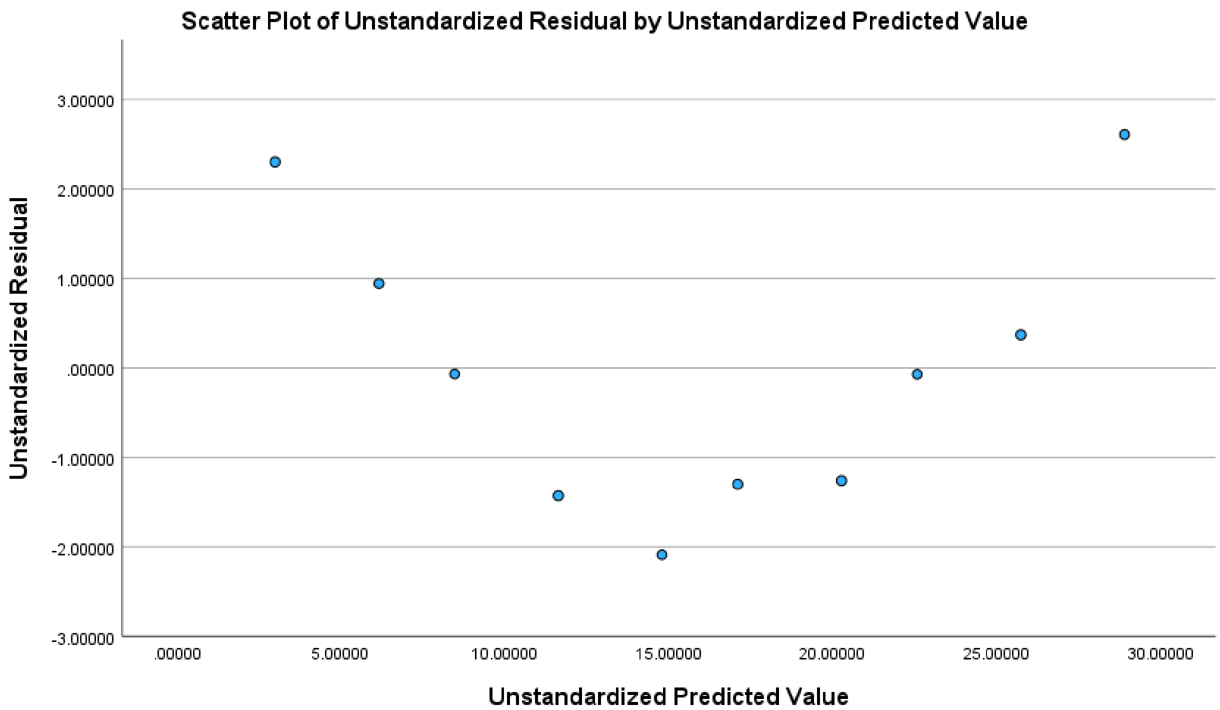
IBM SPSS Vers. 29.0.0.0 will be used for this question.

(a) The beta coefficients are:

Model		Unstandardized Coefficients B	Std. Error	t	Sig.
1	(Constant)	-1.011	3.295	-.307	.768
	X1	2.312	1.315	1.757	.122
	X2	.848	2.083	.407	.696

a. Dependent Variable: Y

(b) The residual plot is



This is clearly not a random scatter, suggesting nonlinear relationships between *Y* and the predictors.

(c) To implement Tukey's test, we fit the model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \gamma \hat{Y}^2$, resulting in:

Model		Unstandardized Coefficients B	Std. Error	t	Sig.
1	(Constant)	5.292	.734	7.209	<.001
	X1	.840	.257	3.265	.017
	X2	-.410	.384	-1.068	.326
	Y_hat_2	.025	.002	14.520	<.001

a. Dependent Variable: Y

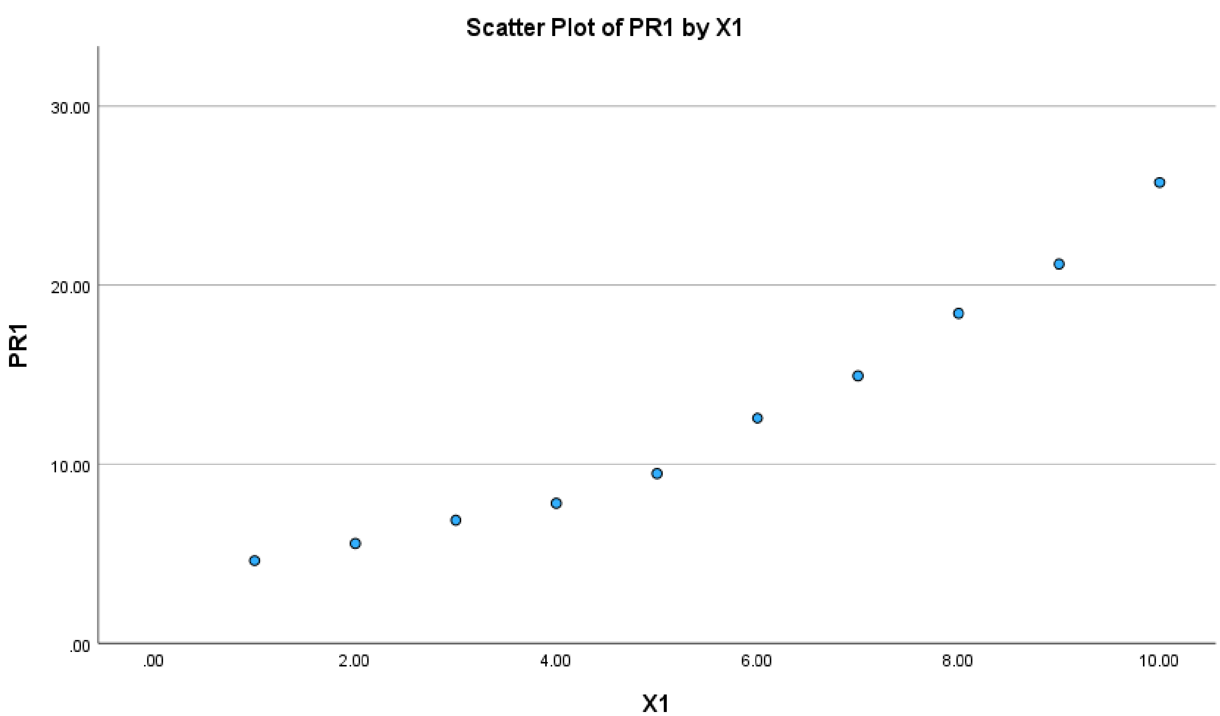
$H_0 : \gamma = 0$ is rejected at the 5% level with a p-value of <.001. Therefore, the relationship between *Y* and the predictors is nonlinear.

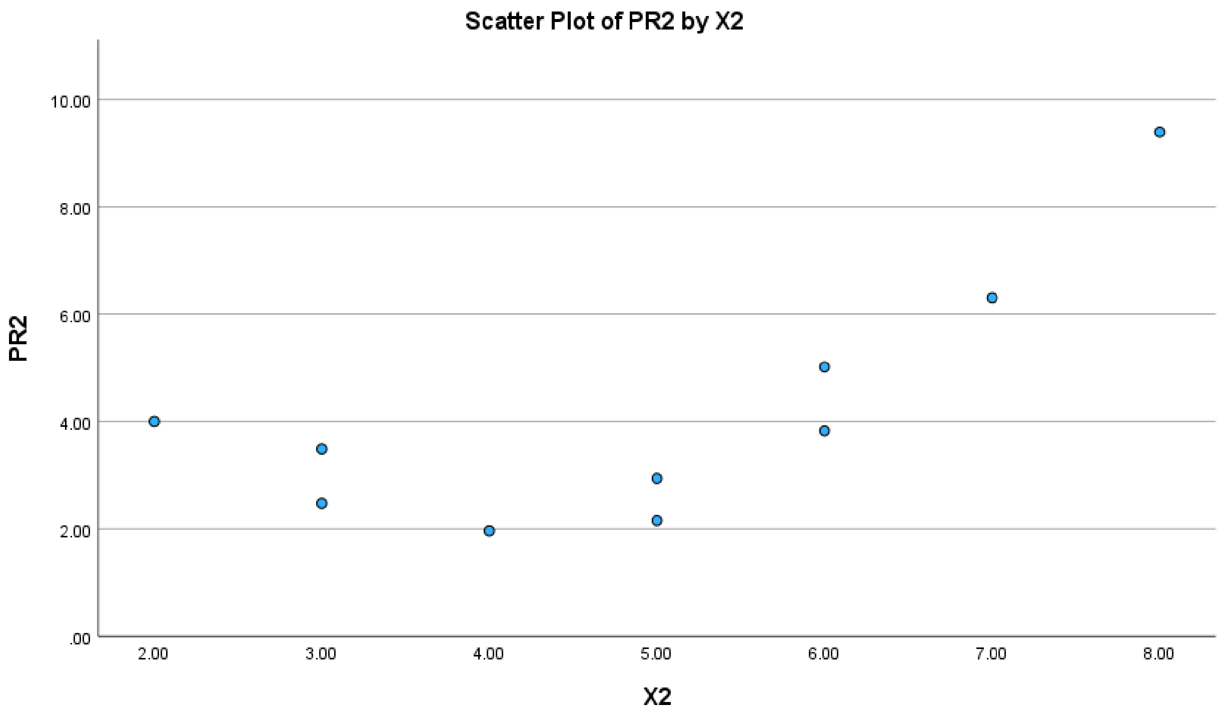
(d) In order to decide which of X_1 and X_2 to square, we produce a partial residual plot for each of the predictors.

For X_1, the partial residuals $PR_1 = \hat{\varepsilon} + \hat{\beta}_1 X_1$ are tabulated below by using the residuals from the model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$ (see (b) above) and remembering that $\hat{\beta}_1 = 2.312$ (see (a) above).

Similarly, for X_2 the partial residuals $PR_2 = \hat{\varepsilon} + \hat{\beta}_2 X_2$ are tabulated below by using the residuals from the model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$ (see (b) above) and remembering that $\hat{\beta}_2 = .848$ (see (a) above).

X1	X2	Y	Residuals	PR1	PR2
1.00	2.00	5.30	2.30258	4.61	4.00
2.00	3.00	7.10	.94258	5.57	3.49
3.00	3.00	8.40	-.06903	6.87	2.47
4.00	4.00	10.20	-1.42903	7.82	1.96
5.00	5.00	12.70	-2.08903	9.47	2.15
6.00	5.00	15.80	-1.30065	12.57	2.94
7.00	6.00	19.00	-1.26065	14.92	3.83
8.00	6.00	22.50	-.07226	18.42	5.02
9.00	7.00	26.10	.36774	21.18	6.30
10.00	8.00	31.50	2.60774	25.73	9.39





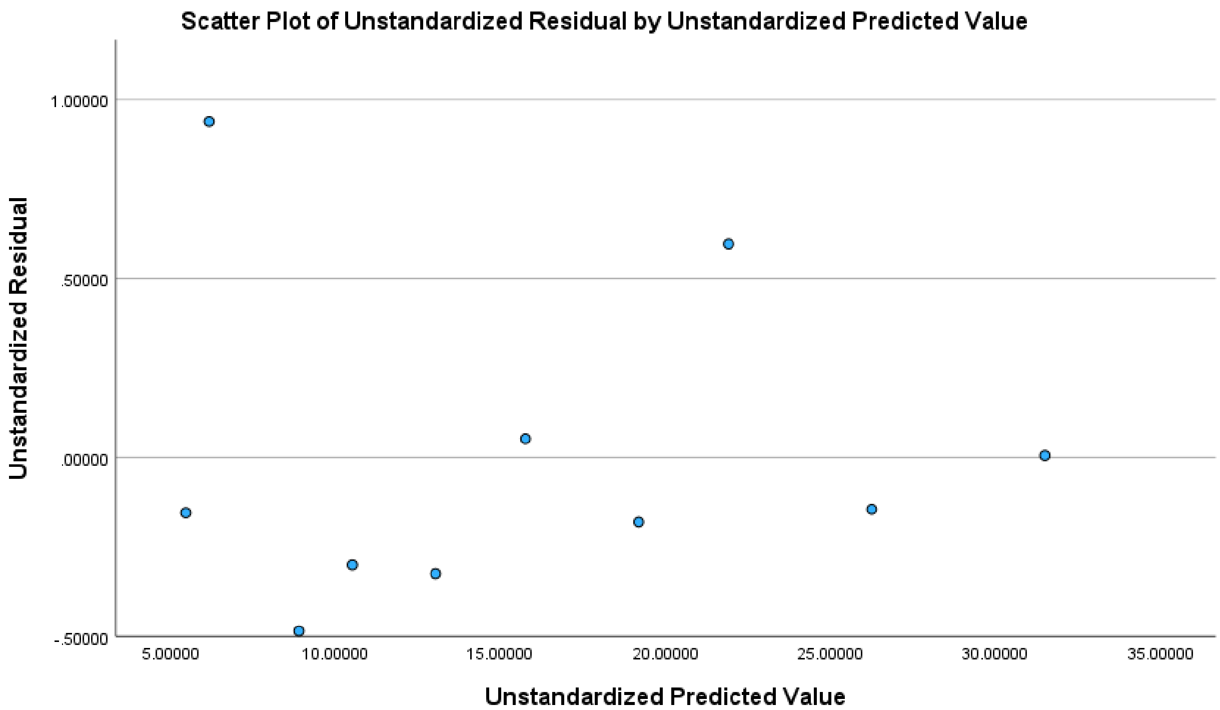
The partial residual plots show that X_1 should enter the model in a linear fashion and X_2 should enter the model in a quadratic fashion. Thus, the model that should be fitted is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_2^2 + \varepsilon$$

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	9.490	1.579		6.011	<.001
	X1	2.723	.396	.942	6.876	<.001
	X2	-4.288	.867	-.937	-4.945	.003
	X2Sq	.454	.053	.997	8.508	<.001

a. Dependent Variable: Y

The residual plot now looks much more random:



(b) Dealing with non-constant variance: Variance-stabilizing transformations

Heterogeneous variance (or heteroscedasticity) occurs when the variance increases (or more rarely, decreases) with $\mathrm{E}(Y \mid \mathbf{X})$. The residual plot then displays a funnel effect. Several possible transformations are possible in order to stabilize the variance

- For Poisson distributed data, $\mathrm{var}(Y \mid \mathbf{X}) \propto \mathrm{E}(Y \mid \mathbf{X})$ and the transformation $\sqrt{Y}$ is usually applied.
- When the standard deviation of *Y* is a fixed proportion of the mean value of *Y*, then $\mathrm{var}(Y \mid \mathbf{X}) \propto \mathrm{E}^2(Y \mid \mathbf{X})$ and the transformation $\log Y$ is usually applied. This is, in fact, one of the most common variance-stabilizing transformations.
- When most of the responses are close to 0 and a few are large, then $\mathrm{var}(Y \mid \mathbf{X}) \propto \mathrm{E}^4(Y \mid \mathbf{X})$ and the transformation $1/Y$ is usually used.
- When the response lies in [0, 1], then the transformation $\sin^{-1}\sqrt{Y}$ is usually used.

The transformations $\sqrt{Y}$, $\log Y$ and $1/Y$ are used, respectively, for mild, moderate and excessive heteroscedasticity.

Example 4.9

Using a statistical software, examine the data below for heteroscedasticity and perform an appropriate transformation to stabilize the variance:

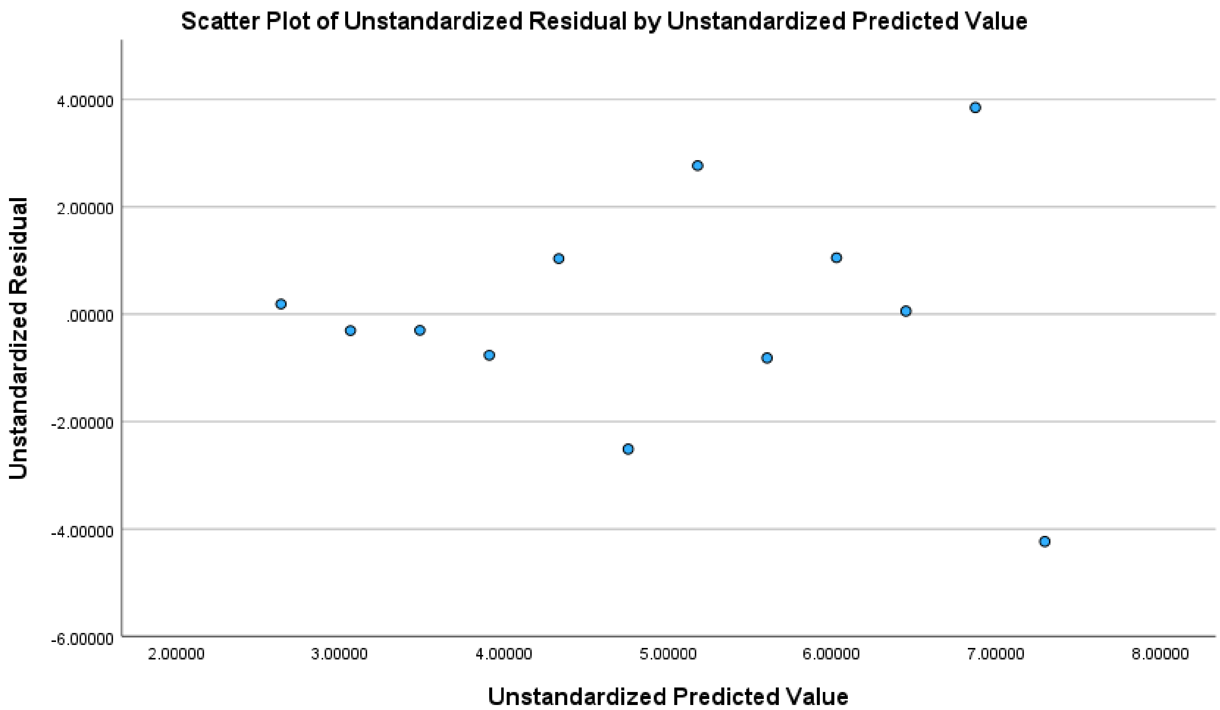
x	y
1	2.825
2	2.755
3	3.183
4	3.142
5	5.365
6	2.238
7	7.943
8	4.782
9	7.074
10	6.501
11	10.717
12	3.056

Solution

Using SPSS Ver 29.0.0.0, we first fit the model $Y = \beta_0 + \beta_1 X + \varepsilon$ and do a residual plot:

Model		Unstandardized Coefficients		t	Sig.
		B	Std. Error		
1	(Constant)	2.214	1.381	1.603	.140
	x	.423	.188	2.255	.048

a. Dependent Variable: y

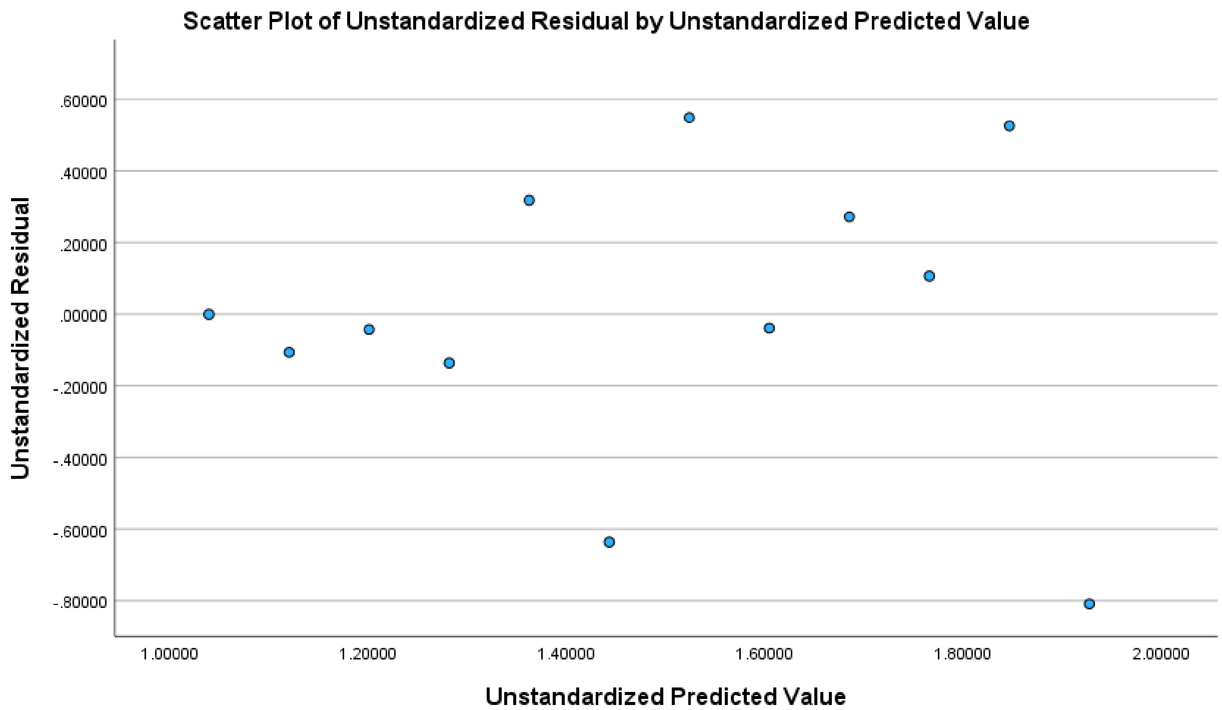


The residual plot shows a funneling out effect. We will try the log transformation and re-do the regression:

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.959	.265		3.613	.005
	x	.081	.036	.578	2.237	.049

a. Dependent Variable: log_y

The residual plot is as follows:



This is more or less a random scatter suggesting the transformation log (*Y*) stabilizes the variance.

4.4 INFLUENCE AND OUTLIERS IN LINEAR REGRESSION

The Hat Matrix

Consider the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Recall that the least-squares estimator of $\boldsymbol{\beta}$ is given by $\hat{\boldsymbol{\beta}} = (\mathbf{X}^{/}\mathbf{X})^{-1}\mathbf{X}^{/}\mathbf{Y}$, so that the fitted values are

$$\begin{aligned}\hat{\mathbf{Y}} &= \mathbf{X}\hat{\boldsymbol{\beta}} \\ &= \mathbf{X}\left\{(\mathbf{X}^{/}\mathbf{X})^{-1}\mathbf{X}^{/}\mathbf{Y}\right\} \\ &= \mathbf{X}(\mathbf{X}^{/}\mathbf{X})^{-1}\mathbf{X}^{/}\mathbf{Y}\end{aligned}$$

The (*n* x *n*) matrix $\mathbf{X}(\mathbf{X}^{/}\mathbf{X})^{-1}\mathbf{X}^{/}$ occurs so often in the literature that we denote it by **H** and call it the hat matrix, i.e.

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^{/}\mathbf{X})^{-1}\mathbf{X}^{/} \qquad \textbf{4-2}$$

so that

$$\hat{\mathbf{Y}} = \mathbf{HY}$$

The last equation explains why **H** is called the hat matrix, it transforms the **Y** vector of observations into the **Y** hat vector of fitted values.

The hat matrix has two interesting properties:

(a) Symmetry, i.e. $\mathbf{H}^{/} = \mathbf{H}$
(b) Idempotence, i.e. $\mathbf{H}^{2} = \mathbf{H}$

Example 4.4

Consider the following data points:

X	.8	.9	1.0	1.1	1.2	4.0
Y	1.5	1.9	1.9	2.0	2.2	13.0

Compute the hat matrix for the data above.

Solution

The **X** matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & .8 \\ 1 & .9 \\ 1 & 1.0 \\ 1 & 1.1 \\ 1 & 1.2 \\ 1 & 4.0 \end{bmatrix}$$

Then, using Mathematica, the hat matrix is

$$\mathbf{H} = \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}' = \begin{bmatrix} .2311 & .2219 & .2127 & .2035 & .1943 & -.0636 \\ .2219 & .2140 & .2061 & .1983 & .1904 & -.0307 \\ .2127 & .2061 & .1996 & .1930 & .1864 & .0022 \\ .2035 & .1983 & .1930 & .1877 & .1825 & .0035 \\ .1943 & .1904 & .1864 & .1825 & .1785 & .0680 \\ -.0636 & -.0307 & .0022 & .0035 & .0680 & .9890 \end{bmatrix}$$

Note that once the hat matrix is given, several calculations are simplified. For example:

(a) *Variance of fitted values*:

$$\begin{aligned} \operatorname{var}\hat{\mathbf{Y}} &= \operatorname{var}\mathbf{HY} \\ &= \mathbf{H}\left(\operatorname{var}\mathbf{Y}\right)\mathbf{H}' \\ &= \mathbf{H}^2\sigma^2 \\ &= \mathbf{H}\sigma^2 \end{aligned}$$

(b) Computation of residuals:

$$\begin{aligned} \hat{\boldsymbol{\varepsilon}} &= \mathbf{Y} - \hat{\mathbf{Y}} \\ &= \mathbf{Y} - \mathbf{HY} \\ &= \left(\mathbf{I} - \mathbf{H}\right)\mathbf{Y} \end{aligned}$$

(c) *Estimation of* σ^2:

$$\begin{aligned} \hat{\sigma}^2 &= \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-(k+1)} \\ &= \frac{\left\{\left(\mathbf{I}-\mathbf{H}\right)\mathbf{Y}\right\}'\left\{\left(\mathbf{I}-\mathbf{H}\right)\mathbf{Y}\right\}}{n-k-1} \\ &= \frac{\mathbf{Y}'\left(\mathbf{I}-\mathbf{H}\right)\mathbf{Y}}{n-k-1} \end{aligned}$$

The diagonals of **H,** namely h_{ii}, will also be of great importance to us. From Eq. (4-2), h_{ii} can be calculated from

$$h_{ii} = \mathbf{x}_i'\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{x}_i ,$$

where $\mathbf{x}_i^{/}$ is the *i*th row of the **X** matrix. It is interesting to note that:

- the sum of the main diagonals of **H** is $k+1$, i.e. $k+1$;
- $1/n \le h_{ii} \le 1$;
- the more distant in the x-direction a point is from the majority of other points, the larger its hat diagonal (h_{ii}) will be.

Measuring Influence

A point (or sometimes a small set of points) is said to be influential if removing it causes a 'substantial' change in the regression (beta) coefficients of the model. That is, if $\hat{\boldsymbol{\beta}}$ is the estimated $\boldsymbol{\beta}$ when all the data are used, and $\hat{\boldsymbol{\beta}}_{(i)}$ is the estimated $\boldsymbol{\beta}$ when the *i*th data point is removed, then an influential point will change the relative magnitude of $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)}$ 'substantially'.

A point may be influential and be a legitimate point (left in Figure 4-1) or may be influential but not belong to the trend set by the balance of the data (right in Figure 4-1). In the latter case, it said to be an outlier (note, however, that not all outliers are influential). It is important to be able to identify both types of points. We usually do not delete influential points unless they are also outliers.

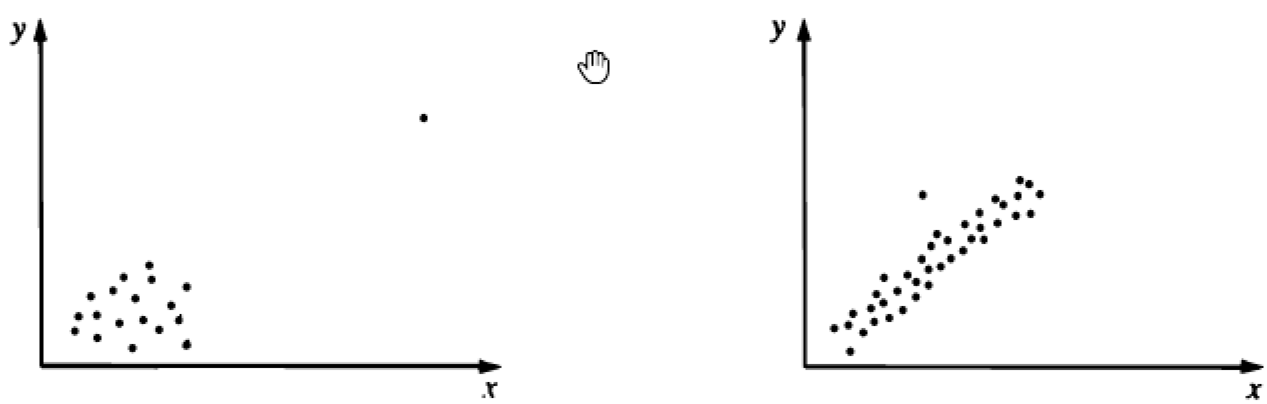


Figure 4-1

The most widely used measure of influence is the Cook's distance, defined for the *i*th point as

$$D_i = \frac{\left(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)}\right)' \left(\mathbf{X}'\mathbf{X}\right)\left(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)}\right)}{(k+1)s^2}$$

For computational purposes we use,

$$D_i = \frac{1}{k+1} r_i^2 \frac{h_{ii}}{1-h_{ii}}, \qquad \textbf{4-3}$$

where

$$r_i = \frac{\hat{\varepsilon}_i}{s\sqrt{1-h_{ii}}}$$

is the standardized residual for the *i*th observation. From Eq. (4-3), it is seen that Cook's distance D_i will be large if a point is off (distant) along the *x*-direction (resulting in a large h_{ii}), if a point is off in the *y*-direction (i.e. is also an outlier), or both. Points with unusually large D_i compared to the majority of other points are the ones which are potentially most influential. A useful cut-off is $D_i > 1$.

The quantities h_{ii}, r_i, D_i and a few others to be introduced later are known as regression diagnostics.

Example 4.5

Consider the data in Example 4.4.

(a) Do a scatterplot of y vs. x
(b) Use a statistical software to find the coefficients of the simple linear regression model $Y = \beta_0 + \beta_1 X + \varepsilon$
(c) Given the matrix

$$\mathbf{H} = \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}' = \begin{bmatrix} .2311 & .2219 & .2127 & .2035 & .1943 & -.0636 \\ .2219 & .2140 & .2061 & .1983 & .1904 & -.0307 \\ .2127 & .2061 & .1996 & .1930 & .1864 & .0022 \\ .2035 & .1983 & .1930 & .1877 & .1825 & .0035 \\ .1943 & .1904 & .1864 & .1825 & .1785 & .0680 \\ -.0636 & -.0307 & .0022 & .0035 & .0680 & .9890 \end{bmatrix}$$

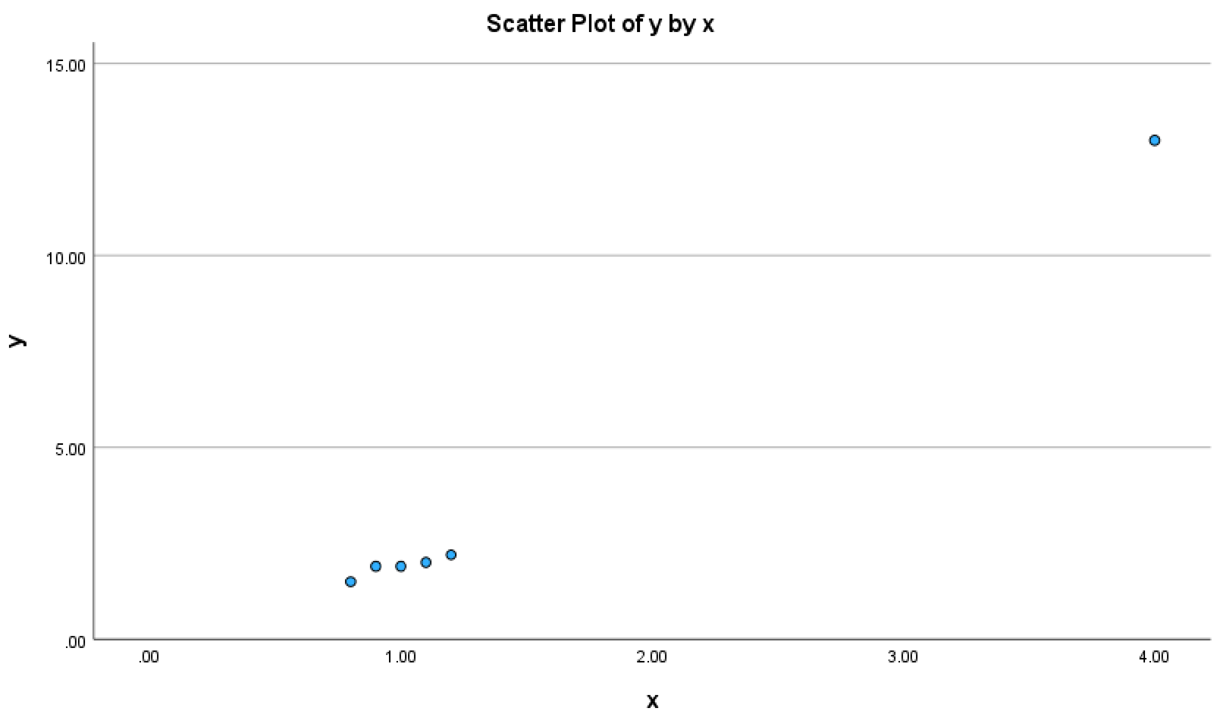
and $MS_{res} = .128$, obtain the hat diagonals, standardized residuals and Cook's distance for each of the 6 data points.

(d) Identify any influential observations and the nature of their influence.

(e) Using statistical software, obtain new least-squares estimates with the influential observations in (d) deleted.

Solution

(a)



(b) From IBM SPSS 29.0.0.0, $\hat{\beta}_0 = -1.757$ and $\hat{\beta}_1 = 3.671$

(c)

x	y	yhat_i	e_i	h_ii	r_i	D_i
0.8	1.5	1.1798	0.32020	0.23110	1.02066	0.15655
0.9	1.9	1.5469	0.35310	0.21400	1.11322	0.16870
1	1.9	1.914	-0.01400	0.19960	-0.04374	0.00024
1.1	2	2.2811	-0.28110	0.18770	-0.87176	0.08780
1.2	2.2	2.6482	-0.44820	0.17850	-1.38218	0.20755
4	13	12.927	0.07300	0.98900	1.94546	170.14401

(d) The sixth data point has an unusually large Cook's distance ($D_i = 170.144$) and is influential. Its hat diagonal is quite large ($h_{ii} = .989$) suggesting that its *x*-distance from most of the other points is a major factor in its influence. Its standardized residual is also somewhat large ($r_i = 1.945$), suggesting its *y*-distance from most points may also play some role and thus it could also be a potential outlier.

(e) With the sixth observation deleted, the new regression coefficients are $\hat{\beta}_0 = .400$ and $\hat{\beta}_1 = 1.500$.

Detecting of outliers

In the last section, we referred to the concept of an outlier and understood it to mean a point that did not belong to the trend set by the balance of the data. It is usually a point that is off in the *y*-direction.

We will now develop a statistical model, called the mean shift outlier model, that will enable us to statistically test for an outlier. Suppose the *i*th observation is suspected to be an outlier. The following multiple regression model is fitted:

$$\mathrm{E}\left(Y \mid x_1, \ldots, x_k\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + \beta_{k+1} U \qquad \textbf{4-4}$$

where

$$U = \begin{cases} 1 & \text{for the } i^{\text{th}} \text{ observation} \\ 0 & \text{for all other observations} \end{cases}$$

It is also assumed that $\mathrm{var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2 \mathbf{I}$

The quantity β_{k+1} is the mean shift in the response for the *i*th observation and a test of $H_0 : \beta_{k+1} = 0$ is a test of whether the *i*th observation is an outlier (if H_0 is rejected then the *i*th data point is indeed an outlier) The test of $H_0 : \beta_{k+1} = 0$ proceeds by creating an indicator variable *U* with is 0 for all observations except the *i*th when it is 1. Thus, in Example 4.4, of the sixth observation is suspected to be an outlier, then the **X** matrix will be defined as follows:

$$\mathbf{X} = \begin{bmatrix} 1 & .9 & 0 \\ 1 & .9 & 0 \\ 1 & 1.0 & 0 \\ 1 & 1.1 & 0 \\ 1 & 1.2 & 0 \\ 1 & 4.0 & 1 \end{bmatrix}$$

Once the model in (4-4) is fitted, the usual *t*-test of $H_0 : \beta_{k+1} = 0$ can be performed:

$$t_{test} = \frac{\hat{\beta}_{k+1}}{\mathrm{SE}\hat{\beta}_{k+1}} \sim t_{n-(k+2)}$$

It can be shown that

$$t_{test} = \frac{\hat{\varepsilon}_i}{\sigma_{(i)}\sqrt{1-h_{ii}}} = r_i\sqrt{\frac{n-(k+2)}{n-(k+1)-r_i^2}},$$

where $\sigma_{(i)}$ the square root of MS_{res} with the i^{th} data point deleted. This formula provides an automatic way of calculating t_{test} without having to use the dummy variable each time.

The quantity t_{test} is also called a Studentized residual.

Example 4.6

Perform an outlier test for the 6th observation in the dataset of Example 4.4. You may use the diagnostics computed in Example 4.5

Solution

The studentized residual for the 6th observation is

$$\begin{aligned} t_{test} &= r_6\sqrt{\frac{n-(k+2)}{n-(k+1)-r_6^2}} \\ &= 1.9455\sqrt{\frac{6-(1+2)}{6-(1+1)-(1.9455)^2}}, \\ &= 7.2668 \end{aligned}$$

Under $H_0: 6^{\text{th}}\text{ observation is not an outlier}$, $t_{test} \sim t_3$ and pval = $2\Pr\{T_3 > 7.2668\}$ lies between .001 and .01. We therefore reject H_0 at the 5th level.

---

5 SELECTING THE BEST MODEL

5.1 EXTRA SUM OF SQUARES

Consider the following two linear regression model:

$$\text{Model A:}\quad \mathrm{E}\left(Y \mid X_1,\ldots,X_k,X_{k+1},\ldots.,X_{k+k'}\right)=\beta_0+\beta_1X_1+\ldots+\beta_kX_k+\beta_{k+1}X_{k+1}+\ldots.+\beta_{k+k'}X_{k+k'}$$

$$\text{Model B:}\quad \mathrm{E}\left(Y \mid X_1,\ldots,X_k\right)=\beta_0+\beta_1X_1+\ldots+\beta_kX_k$$

Since Model B contains the same predictors as Model A up to *k,* Model B is said to be nested within Model A.

Let the regression sum of squares for Model A be written as $SS(\hat{\beta}_0,\hat{\beta}_1,\ldots,\hat{\beta}_{k+k'})$ and that for model B as $SS(\hat{\beta}_0,\hat{\beta}_1,\ldots,\hat{\beta}_k)$. Then the extra sum of squares due to the additional variables $X_{k+1},\ldots,X_{k+k'}$ is

$$SS\left(\hat{\beta}_{k+1},\hat{\beta}_{k+2},\ldots,\hat{\beta}_{k+k'} \mid \hat{\beta}_0,\hat{\beta}_1,\ldots,\hat{\beta}_k\right)=SS\left(\hat{\beta}_0,\hat{\beta}_1,\ldots,\hat{\beta}_{k+k'}\right)-SS\left(\hat{\beta}_0,\hat{\beta}_1,\ldots,\hat{\beta}_k\right)$$

To test the hypothesis $H_0:\beta_{k+1}=\beta_{k+2}=\ldots=\beta_{k+k'}=0$, we use the following test

$$F_{test}=\frac{SS\left(\hat{\beta}_{k+1},\hat{\beta}_{k+2},\ldots,\hat{\beta}_{k+k'} \mid \hat{\beta}_0,\hat{\beta}_1,\ldots,\hat{\beta}_k\right)/k'}{s_A^2}\sim F_{k',n-\left(k+k'+1\right)},$$

where s_A^2 is the mean square error for Model A. If H_0 is rejected then Model A is preferred over Model B. Otherwise, Model B is preferred.

Note that if $k'=1$, a test of $H_0:\beta_{k+1}=0$ can be performed by using a two-tailed *t*-test:

$$t_{test}=\frac{\hat{\beta}_{k+1}}{\mathrm{SE}\hat{\beta}_{k+1}}\sim t_{n-(k+2)}$$

The *F*-test in this case is called a partial *F*-test and is simply t_{test}^2: both tests result in exactly the same p-value.

Example 5.1

The data below shows the extent to which the employees of a form are committed to following a physical exercise regimen (‘exercise_commitment’). The predictors used were:

- diet_intensity: degree to which employee dieted
- social_affiliation_needs: need for social affiliation
- selfacceptance: employee’s self-acceptance
- selfesteem: employee’s self-esteem
- bodyesteem: esteem emplueee felt about their body

CaseID	exercise_commitment	diet_intensity	social_affiliation_needs	selfacceptance	selfesteem	bodyesteem
1	3.43	2.88	99	59	33	57
2	2.29	3.25	104	55	42	41
3	1.71	2.88	92	68	42	31
4	3.29	2.63	87	61	34	54
5	3.71	3.25	112	71	43	59
6	3.29	3.38	101	59	40	54
7	3.71	4.63	111	49	37	61
8	1.43	1.38	100	62	44	26
9	2.43	2.00	109	58	44	42
10	3.86	4.25	97	72	42	62
11	5.00	5.00	101	68	33	79
12	2.43	2.75	109	70	40	41
13	4.00	4.00	94	67	28	64
14	2.86	3.50	81	48	35	49
15	5.00	3.88	96	71	46	81
16	2.43	1.50	89	65	27	41
17	2.43	2.50	104	62	40	39
18	3.14	2.25	106	62	38	50
19	3.29	3.38	97	59	29	54
20	3.14	3.00	93	54	34	52
21	3.43	3.00	110	58	30	56
22	1.86	2.38	78	57	32	31
23	1.71	2.88	104	69	40	30
24	2.71	1.88	76	45	32	46
25	1.71	3.75	106	58	42	28
26	2.14	3.50	93	53	39	37
27	2.14	2.38	97	61	49	38
28	3.86	3.88	75	67	40	65
29	5.00	4.50	90	63	41	79
30	3.29	2.00	114	62	43	56

Using a statistical software:

(a) fit a multiple linear regression of the outcome on all 5 predictors. Shoe the ANOVA table, the table of beta coefficients

(b) is the original model significantly different from the model containing only significant predictors?

Solution

(a) The ANOVA table shows that there is a significant regression effect (pval <.001):

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	28.196	5	5.639	1012.745	<.001[b]
	Residual	.134	24	.006		
	Total	28.330	29			

a. Dependent Variable: exercise_commitment

b. Predictors: (Constant), bodyesteem, social_affiliation_needs, selfacceptance, selfesteem, diet_intensity

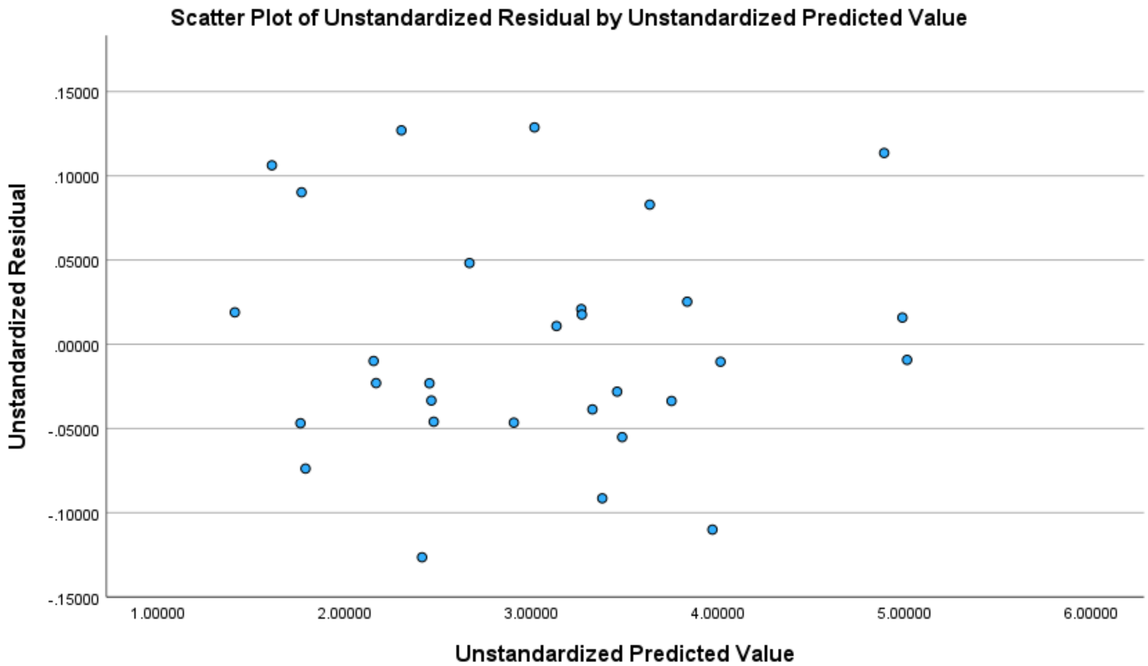
The table of beta coefficients shows that only 'selfesteem' (pval=.044) and 'bodyesteem' (pval<.001) at significant at the 5% level.

Coefficients[a]

Model		Unstandardized Coefficients		t	Sig.
		B	Std. Error		
1	(Constant)	-.445	.166	-2.683	.013
	diet_intensity	.026	.020	1.324	.198
	social_affiliation_needs	.002	.001	1.312	.202
	selfacceptance	.004	.002	1.691	.104
	selfesteem	-.006	.003	-2.124	.044
	bodyesteem	.064	.001	50.642	<.001

a. Dependent Variable: exercise_commitment

The residual plot shows a random cloud:



(b) The model with only significant predictors contained only 'selfesteem' and 'bodyesteem':

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	28.158	2	14.079	2205.135	<.001[b]
	Residual	.172	27	.006		
	Total	28.330	29			

a. Dependent Variable: exercise_commitment

b. Predictors: (Constant), bodyesteem, selfesteem

Coefficients[a]

Model		Unstandardized Coefficients		t	Sig.
		B	Std. Error		
1	(Constant)	-.149	.119	-1.259	.219
	selfesteem	-.003	.003	-1.107	.278
	bodyesteem	.066	.001	65.695	<.001

a. Dependent Variable: exercise_commitment

The extra sum of squares due to ‘diet_intensity’, ‘social_affiliation_needs’ and ‘selfacceptance’ is

$$SS\left(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3 \mid \hat{\beta}_4, \hat{\beta}_5\right) = 28.196 - 28.158$$
$$= .038$$

Under $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$,

$$\begin{aligned} F_{test} &= \frac{SS\left(\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3 \mid \hat{\beta}_4, \hat{\beta}_5\right)/3}{\sigma^2} \\ &= \frac{.038/3}{.006} \\ &= 2.111 \\ &\sim F_{3,24} \end{aligned}$$

The p-value is $\Pr\{F_{3,24} > 2.111\} = .12534$, so we cannot reject H_0 at the 5% level. Therefore, the simpler model is significantly different than the full model and is preferred.

5.2 THE AKAIKE INFORMATION CRITERION (AIC) AND BAYESIAN INFORMATION CRITERION (BIC)

The extra-sum-of-squares method can be applied only when one moder is nested into another. A more model for comparing any two regression models makes use of the Akaike Information Criterion (*AIC*) and Bayesian Information Criterion (*BIC*)

The Akaike Information Criterion (*AIC*), defined by

$$AIC_A = -2\log L + 2\left(k + 2\right),$$

where L is the maximized likelihood of the model and *k* is the number of predictors. As the fit gets better, L increases and AIC_A becomes smaller (because of the negative sign in front of L). Since a fit could be artificially increased simply by adding redundant predictors, a penalty of *k* + 2 is added to AIC_A (*k* + 1 for number of beta parameters, and an extra 1 for the estimated variance).

Consider now a second linear regression model:

Model B: $\mathrm{E}\left(Y \mid X_1, ..., X_k, X_{k+1}, ..., X_{k+k'}\right) = \beta_0 + \beta_1 X_1 + ... + \beta_k X_k + \beta_{k+1} X_{k+1} + ... + \beta_{k+k'} X_{k+k'}$

with Akaike Information Criterion AIC_B.

If $AIC_A \leq AIC_B$, then model A is preferred on the basis of predictive ability. Another measure that is more targeted towards finding the more parsimonious model since it imposes a stronger penalty for complexity (i.e. the number of predictors) is the Bayesian Information Criterion (*BIC*), defined for model A as

$$BIC_A = -2\log L + (k+2)\log n$$

If $BIC_A \leq BIC_B$, then model A is preferred on the basis of parsimony (simplicity).

The BIC tends to be harsher than the *AIC* and it requires larger numbers to be convincing. For example, if $AIC_A \leq AIC_B$ and $AIC_B - AIC_A < 2$, then Model A is preferred but Model B is almost the same in terms of predictive ability. If $AIC_B - AIC_A > 10$, the model with the smaller *AIC* is preferred and the other is much worse in terms of predictive ability. On the other hand, if $BIC_A \leq BIC_B$ and $BIC_B - BIC_A < 2$, then Model A is preferred and slightly better than Model B. If $BIC_B - BIC_A > 10$, the Model B is preferred and the other is somewhat worse in terms of predictive ability.

Example 5.2

In Example 5.1, obtain the AIC's amd BIC's for each of the models in (a) and (b), and indicate your preference for either model.

Solution.

Model (a) uses all 5 predictors: 'diet_intensity', 'social_affiliation_needs', 'selfacceptance', 'selfesteem', and 'bodyesteem'. From the 'Generalized Linear Models' option in SPSS 29.0.0.0., *AIC(a)* = -63.278 and *BIC(b)*= -53.470.

Model (b) uses only 'selfesteem' and 'bodyesteem', *AIC(b)* = -61.641 and *BIC(b)*= -56.036.

Thus, if the aim is predictive ability, model (a) is preferred; if the aim is model simplicity, model (b) is preferred.

5.3 STEPWISE METHODS FOR FINDING THE BEST MODEL

Consider an outcome *Y* and *k* available predictors, where *k* is usually a large number (from 20 to potentially 1000)

<u>Best Subset Selection</u> A useful method for find the best possible model is do all possible regressions (there is $2^k - 1$ possible regressions) and then select the 'best' (either the one with the smallest AIC, or largest R^2 or largest R^2_{adj}). A software will usually be able to give the *p* best regression models (where $p < k$ for a *p* specified by the user).

The best-subset selection procedure becomes increasingly inconvenient for larger values of *k* (e.g. *k* > 30).

<u>Forward Selection</u> In this method, all *k* predictor are regressed on *Y* <u>one at a time</u>. Of all predictors that are significant at an α-level, the most significant one is included in the model. Say this predictor is X_1. If none are found, the procedure stops and the best model is $\hat{Y} = \overline{Y}$. Otherwise, given X_1 is already in the model, the next most significant predictor (obtained using a partial *F*-test) with a *p*-value < α is chosen. Say this is X_2. Given X_1 and X_2 are already is in the model, the next most significant predictor with a *p*-value < α is chosen. The procedure continues until no predictors which are significant at α can be found. Usually, $\alpha = .05$

<u>Backward Elimination</u> In this method, all *k* predictors are fitted in the model. Of all predictors that are not significant at an α-level, the least significant one is removed. Say this predictor is X_1. If none are found, the procedure stops and the best model is $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + ... + \hat{\beta}_k X_k + \varepsilon$. Otherwise, given $X_2, ..., X_k$ are now in the model, the next least significant predictor with a *p*-value $\geq \alpha$ is removed. Say this is X_2. Given $X_3, ..., X_k$ are now is in the model, the next least significant predictor with a *p*-value $\geq \alpha$ is removed. The procedure continues until no predictors with a *p*-value $\geq \alpha$ can be removed. Usually, $\alpha = .05$

<u>Stepwise Regression</u> This method starts like the forward selection procedure - all *k* predictor are regressed on *Y* <u>one at a time</u>, and of all predictors that are significant at an α-level, the most significant one is included in the model. Say this predictor is X_1. If none are found, the procedure stops and the best model is $\hat{Y} = \overline{Y}$. Otherwise, given X_1 is already in the model, the next most significant predictor (obtained using a partial *F*-test) with a *p*-value < α_1 is chosen. Say this is X_2. Given X_1 and X_2 are already is in the model, we test each of $H_0 : \beta_1 = 0$ and $\beta_2 = 0$ separately. We then remove the least significant predictor that has a p-value $\geq \alpha_2$. We keep on adding and removing predictors until no such predictors can be found. Usually, $\alpha_1 = .05$ and $\alpha_2 = .05$.

Example 5.3

In Example 5.1, using a statistical software, run a:

(a) Best-subset regression based on the largest R^2_{adj}
(b) Forward selection
(c) Backward elimination
(d) Stepwise regression

Solution

(a) The top 10 models together with their associated R^2_{adj} are:

		Model									
		1	2	3	4	5	6	7	8	9	10
Adjusted R Square		.994	.994	.994	.994	.994	.994	.994	.994	.994	.993
Effect	diet_intensity	✓	✓			✓		✓		✓	✓
	social_affiliation_needs	✓		✓		✓	✓				✓
	selfacceptance	✓	✓	✓	✓			✓	✓		✓
	selfesteem	✓	✓	✓	✓	✓	✓			✓	
	bodyesteem	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

The model building method is Best Subsets using the Adjusted R Square criterion.
A checkmark means the effect is in the model.

The best model is the one that includes all five predictors

Model Term	Coefficient ▼	Std.Error	t	Sig.	95% Confidence Interval		Importance
					Lower	Upper	
Intercept	-0.445	0.166	-2.683	.013	-0.788	-0.103	
bodyesteem	0.064	0.001	50.642	.000	0.061	0.067	0.996
selfesteem	-0.006	0.003	-2.124	.044	-0.011	-0.000	0.002
selfacceptance	0.004	0.002	1.691	.104	-0.001	0.008	0.001
diet_intensity	0.026	0.020	1.324	.198	-0.015	0.067	0.001
social_affiliation_needs	0.002	0.001	1.312	.202	-0.001	0.005	0.001

(b) Using $\alpha = .05$, the best model using forward selection is:

Coefficients[a]

Model		Unstandardized Coefficients		t	Sig.
		B	Std. Error		
1	(Constant)	-.267	.052	-5.155	<.001
	bodyesteem	.066	.001	66.136	<.001

a. Dependent Variable: exercise_commitment

(c) Using $\alpha = .05$, the best model using backward elimination is the same as in (b).

(d) Using $\alpha_1 = .05$ and $\alpha_2 = .051$, the best model using stepwise regression is the same as I (b).

6 LOGISTIC REGRESSION

6.1 BERNOULLI DISTRIBUTION

If we have a series of outcomes each of which can take two values ('disease' vs.'healthy', 'success' vs. 'failure', etc.), then each outcome is said to be a Bernoulli random variable. Let the *i*th outcome be denoted by Y_i. Then we may write

$$Y_i = \begin{cases} 1 & \text{if the } i^{\text{th}} \text{ outcome is a 'success' (occuring with probability } p) \\ 0 & \text{if the } i^{\text{th}} \text{ outcome is a 'failure' (occuring with probability } q = 1-p) \end{cases}$$

Since $Y_i = 1$ denotes success and *p* is the probability of success, then $p = \Pr\{Y_i = 1\}$.

'Success' here should <u>not</u> be interpreted as a positive event. In fact, it is the event of interest. For example, if we wish to find predictors of 'disease', then 'disease' is our success.

Any Y_i defined as above is said to be a Bernoulli (*p*) random variable (r.v.). Note that

$$\mathrm{E}Y_i = p,\ \mathrm{var}Y_i = p(1-p)$$

6.2 BINARY LOGISTIC REGRESSION

If Y is binary, and if we want to obtain the predictors of success ($Y = 1$), then we cannot use linear regression since the latter assumes a normally distributed outcome. We appropriate model to run is a <u>binary logistic regression</u> defined by

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + ... + \beta_k X_k, \qquad \textbf{6-1}$$

where $X_1, ..., X_k$ are predictors of success.

The quantity $p/(1-p)$ is extremely important in interpreting the results of a logistic regression and is called the <u>odds of success,</u> i.e.

$$\text{Odds (success)} = \frac{p}{1-p} \qquad \textbf{6-2}$$

Thus, if the probability of an event is 1/5, then its odds (the odds in favor of it) are (1/5)/(4/5) = 1 to 4. On the other hand, odds of 3 to 1 mean a probability of success of

$$p = \frac{\text{Odds}}{1+\ \text{Odds}} = \frac{3}{1+3} = \frac{3}{4}$$

Once the model is fitted,

$$\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + ... + \hat{\beta}_k X_k,$$

Given our knowledge of odds, we are now able to write the above as

$$\log\left(\text{Odds}\right)_{(X_1,...,X_k)} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + ... + \hat{\beta}_k X_k.$$

If X_1 is increased to $X_1 + 1$ while keeping all other X_j's constant, then

$$\log\left(\text{Odds}\right)_{(X_1+1,...,X_k)} = \hat{\beta}_0 + \hat{\beta}_1\left(X_1 + 1\right) + ... + \hat{\beta}_k X_k.$$

Subtracting the last two equations,

$$\hat{\beta}_1 = \log\left(\text{Odds}\right)_{X_1+1} - \log\left(\text{Odds}\right)_{X_1} = \log\frac{\left(\text{Odds}\right)_{X_1+1}}{\left(\text{Odds}\right)_{X_1}}$$

$$\boxed{\therefore \quad \exp\hat{\beta}_1 = \frac{\left(\text{Odds}\right)_{X_1+1}}{\left(\text{Odds}\right)_{X_1}} = OR} \qquad \textbf{6-3}$$

since the ratio of two odds is an odds ratio (*OR*). A similar relationship holds for all other $\hat{\beta}_j$'s.

Eq. (6-3) is used in the following manner once the binary logistic regression model is fitted:

(1) <u>The case X_1 is continuous</u>. *The estimated value $\hat{\beta}_1$ is exponentiated and is interpreted as the factor by which the odds of success change for each unit increase in X_1, keeping all other X_j's constant, i.e. $\exp\hat{\beta}_1$ is the odds ratio (OR) associated with X_1.* Similarly for all continuous X_j's

(2) <u>The case X_1 is dichotomous</u>. *The estimated value $\hat{\beta}_1$ is exponentiated and is interpreted as the factor by which the odds of success change from $X_1 = 0$ to $X_1 = 1$, keeping all X_j's constant, i.e. $\exp \hat{\beta}_1$ is the odds ratio (OR) associated with X_1. Similarly for all dichotomous X_j's.*

Example 6.1

The data below show the pain status (*Y* = 1: no pain, *Y* = 0: pain) measured after 6 weeks of treatment of each of 18 subjects together to their treatment group ($X_1 = 1 : \text{treatment}$, $X_1 = 0 : \text{control}$), their age (X_2), sex ($X_3 = 1 : \text{female}, X_3 = 0 : \text{male}$), and pre-treatment duration of symptoms (X_4).

Y	X1	X2	X3	X4
1	1	76	0	36
1	1	70	0	22
0	1	80	1	33
1	1	77	0	33
0	1	73	1	17
1	1	84	1	84
1	1	71	0	24
1	1	78	1	96
1	1	83	1	61
0	0	75	1	60
0	0	78	0	8
0	0	82	1	35
0	0	80	1	3
0	0	75	1	27
0	0	80	0	60
1	0	71	1	8
0	0	83	1	5
0	0	70	1	26

Fit a logistic model to the data and find the significant predictors of pain absence.

Solution

The significance of the effects are:

Tests of Model Effects

Source	Type III Wald Chi-Square	df	Sig.
(Intercept)	.967	1	.326
X1	4.074	1	.044
X3	.775	1	.379
Age	1.067	1	.302
Duration	1.243	1	.265

Dependent Variable: Y
Model: (Intercept), X1, X3, Age, Duration

The table of odds ratios are:

Parameter Estimates

			95% Wald Confidence Interval		Hypothesis Test				95% Wald Confidence Interval for Exp(B)	
Parameter	B	Std. Error	Lower	Upper	Wald Chi-Square	df	Sig.	Exp(B)	Lower	Upper
(Intercept)	15.431	16.3877	-16.688	47.551	.887	1	.346	5031989.379	5.656E-8	4.476E+20
[X1=1]	3.163	1.5672	.092	6.235	4.074	1	.044	23.647	1.096	510.274
[X1=0]	0[a]	.	.	.	.	.	.	1	.	.
[X3=1]	-1.373	1.5601	-4.431	1.684	.775	1	.379	.253	.012	5.389
[X3=0]	0[a]	.	.	.	.	.	.	1	.	.
Age	-.239	.2319	-.694	.215	1.067	1	.302	.787	.500	1.240
Duration	.049	.0437	-.037	.134	1.243	1	.265	1.050	.964	1.144
(Scale)	1[b]									

Dependent Variable: Y
Model: (Intercept), X1, X3, Age, Duration
a. Set to zero because this parameter is redundant.
b. Fixed at the displayed value.

The only significant predictor is group – compared to subjects in the control group, those in the treatment group have higher odds of pain absence by a factor of 23.647 [OR=23.647, 95% CI = (1.096, 510.274), pval=.044].

Most of the diagnostics and methods of multiple linear regression can be used in logistic regression, e.g. Cook's distance, hat diagonals, Studentized residuals, and stepwise methods. However:

- instead of R^2 or R^2_{adj}, the Nagelkerke R^2 is used to represent the amount of variation in Y explained by the predictors. For the model in Example 6.1, the Nagelkerke R^2 is 64.2%.

- Instead of the residual sum of squares, we use the deviance, defined by

$$G^2 = -2\sum_{i=1}^{n}\left\{y_i \log \hat{p}_i + \left(1-y_i\right)\log\left(1-\hat{p}_i\right)\right\},$$

where $\hat{p}_i$ the fitted probability for the ith observation.

6.3 BINOMIAL LOGISTIC REGRESSION

Sometimes, data for logistic regression are given in grouped form. For example, for the first row of the data in Example 6.1, the response on one subject with $X_1 = 1,\ X_2 = 76,\ X_3 = 0,$ and $X_4 = 36$ was taken as *Y* = 1.

Suppose, instead, for the same combination of X_i's, the response on n_1 subjects were taken and Y_1 of them were pain free. Similarly for the other rows, then our data could look as follows:

X1	X2	X3	X4	n_i	y_i
1	76	0	36	10	7
1	70	0	22	8	5
1	80	1	33	12	2
1	77	0	33	11	7
1	73	1	17	5	0
1	84	1	84	15	13
1	71	0	24	10	6
1	78	1	96	11	9
1	83	1	61	12	10
0	75	1	60	7	2
0	78	0	8	6	1
0	82	1	35	12	2
0	80	1	3	10	3
0	75	1	27	9	0
0	80	0	60	8	0
0	71	1	8	9	1
0	83	1	5	10	2
0	70	1	26	12	3

This data can still be analyzed by using logistic regression – this time, it is called binomial logistic regression. The word binomial is used because for each combination of X_i's, we have Y_i .'successes' out of n_i trials. We assume that $Y_i \sim \text{bino}(n_i, p_i)$. The binomial logistic regression proceeds in the same way as the binary logistic regression except that the outcome now is the pair (Y_i, n_i)

Example 6.2

Fit a logistic model to the binomial data above and find the significant predictors of pain absence.

Solution

The significance of the effects are:

Tests of Model Effects

Source	Type III		
	Wald Chi-Square	df	Sig.
(Intercept)	.500	1	.479
Group	15.668	1	<.001
Sex	.768	1	.381
Age	.112	1	.738
Duration	5.804	1	.016

Events: y_i
Trials: n_i
Model: (Intercept), Group, Sex, Age, Duration

The table of odds ratios are:

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval		Hypothesis Test			Exp(B)	95% Wald Confidence Interval for Exp(B)	
			Lower	Upper	Wald Chi-Square	df	Sig.		Lower	Upper
(Intercept)	-3.102	3.4596	-9.883	3.678	.804	1	.370	.045	5.103E-5	39.584
[Group=1]	1.615	.4080	.815	2.415	15.668	1	<.001	5.027	2.260	11.184
[Group=0]	0[a]	.	.	.	.	.	.	1	.	.
[Sex=1]	-.368	.4198	-1.191	.455	.768	1	.381	.692	.304	1.576
[Sex=0]	0[a]	.	.	.	.	.	.	1	.	.
Age	.016	.0471	-.077	.108	.112	1	.738	1.016	.926	1.114
Duration	.021	.0086	.004	.038	5.804	1	.016	1.021	1.004	1.038
(Scale)	1[b]									

Events: y_i
Trials: n_i
Model: (Intercept), Group, Sex, Age, Duration

a. Set to zero because this parameter is redundant.

b. Fixed at the displayed value.

There are two significant predictors:

- Group: compared to the control group, the treatment group have significantly higher odds of pain absence by a factor of 5.027 [OR=5.027, 95% CI = (2.260, 11.184), pval <.001].
- Duration: For each unit increase in duration, the odds of pain absence increase significantly by factor of 1.021 [OR=1.021, 95% CI = (1.004, 1.038), pval = .016].